

The Baker–Campbell–Hausdorff Formula: Complete Proofs, All-Order Expansions, and Analytic Qualifications

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Abstract

We give a systematic proof of the mathematical assertions and displayed identities in the Wikipedia article on the Baker–Campbell–Hausdorff formula, with necessary corrections to its geometric and operator-theoretic qualifications. The treatment separates formal identities, convergent local identities, and global exponential identities. A direct proof of the Dynkin–Specht–Wever projection proves Friedrichs’ characterization of primitive series and yields Dynkin’s formula with every coefficient specified. An independent differential proof produces the Bernoulli recursion and the Poincaré integral formula. Explicit finite word sums describe every homogeneous and bihomogeneous BCH component. For the Zassenhaus product we prove existence, uniqueness, two recursive constructions, a nonrecursive weighted-tree formula, and a quantitative convergence criterion. We also prove the local Lie correspondence, product formulas of Trotter and Suzuki, all-order Maurer–Cartan and metric expansions, and the quantum-mechanical identities in concrete Hilbert-space realizations. Counterexamples clarify logarithm branches, nonconvergence, degenerate pullbacks, and the distinction between infinitesimal and exponentiated commutation relations. Exact rational computations through degree ten accompany the proofs.

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1 Scope, conventions, and the three meanings of BCH

The central problem is to describe a logarithm of a product of exponentials:

$$e^X e^Y = e^{Z(X,Y)}. \quad (1.1)$$

The reference page is the revision of the Wikipedia article *Baker–Campbell–Hausdorff formula* with revision identifier 1368909113, last edited on 11 August 2026 and consulted on 16 September 2026 [1]. The coverage table in [section D](#) identifies where each family of assertions is proved. Historical attributions and lists of links are not mathematical theorems; likewise, a theorem merely named as an application is not thereby stated on the page. In particular, the Stone–von Neumann classification theorem and the Golden–Thompson inequality are not additional claims that the present paper purports to prove.

There are three distinct settings.

Formal algebra. X, Y have positive degree. Every identity is an identity of formal series, and every fixed-degree coefficient is a finite sum. No numerical convergence is intended.

Local analysis. X, Y belong to a Banach algebra or a finite-dimensional Lie algebra and are sufficiently small. The series converges and specifies the logarithm branch issuing from zero.

Global identities under special relations. Commutation relations may allow the series to terminate or to be replaced by a closed expression. An independent differential argument can then prove an exponential identity beyond the convergence domain of its Taylor series.

Conflating these settings is the source of several apparent paradoxes. A matrix may have logarithms even though its BCH series diverges; a closed expression may continue the local BCH logarithm without making the original series convergent; a logarithm need not belong to the Lie algebra generated by its two inputs.

Throughout, $[A, B] = AB - BA$ in an associative algebra and $\text{ad}_A B = [A, B]$. Composition of adjoint maps is read from right to left. For a nonempty word $w = a_1 \cdots a_n$ define its *right-nested bracket* by

$$[a_1 \cdots a_n]_{\text{R}} = [a_1, [a_2, \dots, [a_{n-1}, a_n] \dots]], \quad [a]_{\text{R}} = a. \quad (1.2)$$

Thus $[XXY]_{\text{R}} = [X, [X, Y]]$; a superscript inside a word denotes repeated letters, not an associative power after the bracketing operation has been performed. We write Z_n for total degree n , and $Z_{p,q}$ for degree p in X and degree q in Y .

The scalar functions that organize the entire subject are

$$\phi_{-}(z) = \frac{1 - e^{-z}}{z}, \quad \phi_{+}(z) = \frac{e^z - 1}{z}, \quad f(z) = \frac{z}{1 - e^{-z}}, \quad \psi(x) = \frac{x \log x}{x - 1}. \quad (1.3)$$

The first two are entire, with value 1 at zero. The function f is meromorphic with a removable singularity at zero and poles at the nonzero points $2\pi i\mathbb{Z}$. The function ψ is used near $x = 1$, with a logarithm branch analytic there. Expressions such as $\phi_{-}(M)$ always denote functional calculus, not an assumption that M is invertible.

2 Formal series and a coefficient formula for every word

2.1 The completed associative algebra

Let $\mathbb{T}$ be a field of characteristic zero and $V = \mathbb{T}X \oplus \mathbb{T}Y$. The tensor algebra and its degree completion are

$$T(V) = \bigoplus_{n \geq 0} V^{\otimes n}, \quad \widehat{T}(V) = \prod_{n \geq 0} V^{\otimes n}.$$

Multiplication is concatenation, extended by the Cauchy product. The filtration $F^m \widehat{T} = \prod_{n \geq m} V^{\otimes n}$ satisfies $F^m F^n \subseteq F^{m+n}$. Consequently, substitution of positive-degree series into another formal series is well defined.

Lemma 2.1 (Formal exponential and logarithm). *The maps*

$$\exp A = \sum_{n \geq 0} \frac{A^n}{n!}, \quad \log(1 + U) = \sum_{k \geq 1} \frac{(-1)^{k-1}}{k} U^k \quad (2.1)$$

are mutually inverse bijections between $F^1 \widehat{T}$ and $1 + F^1 \widehat{T}$. If A and B commute, then $e^{A+B} = e^A e^B$.

Proof. The series are defined because only finitely many powers contribute in any fixed degree. The scalar formal identities $\log(e^t) = t$ and $e^{\log(1+t)} = 1 + t$ follow, for example, by formal differentiation and comparison of constant terms. Substituting A or U is legitimate: powers of one fixed algebra element commute with each other, regardless of whether the ambient algebra is commutative. If $AB = BA$, the binomial theorem gives $(A + B)^n = \sum_{j=0}^n \binom{n}{j} A^j B^{n-j}$; summing proves the last assertion. No commutativity of unrelated elements has been used. $\square$

Define

$$Z(X, Y) = \text{BCH}(X, Y) = \log(e^X e^Y) = \sum_{n \geq 1} Z_n(X, Y). \quad (2.2)$$

By lemma 2.1, this is the unique positive-degree formal solution of (1.1). Expanding the two exponentials first gives

$$Z(X, Y) = \sum_{k \geq 1} \frac{(-1)^{k-1}}{k} \sum_{\substack{r_i, s_i \geq 0 \\ r_i + s_i > 0 \ (1 \leq i \leq k)}} \frac{X^{r_1} Y^{s_1} \dots X^{r_k} Y^{s_k}}{\prod_{i=1}^k r_i! s_i!}. \quad (2.3)$$

This proves the associative expansion on the reference page. It does not yet prove that its homogeneous parts are Lie polynomials.

2.2 A finite nonrecursive formula

For a nonempty word w in X, Y , put

$$a(w) = \begin{cases} (r!s!)^{-1}, & w = X^r Y^s, \quad r + s = |w|, \\ 0, & \text{otherwise.} \end{cases} \quad (2.4)$$

Equivalently, $a(w) = 0$ precisely when w contains the adjacent pattern YX . Pure powers are permitted.

Theorem 2.2 (Every associative BCH coefficient). *For $w = a_1 \cdots a_n$, the coefficient $c(w)$ of w in Z is*

$$c(w) = \sum_{k=1}^n \frac{(-1)^{k-1}}{k} \sum_{0=i_0 < i_1 < \cdots < i_k=n} \prod_{j=1}^k a(a_{i_{j-1}+1} \cdots a_{i_j}). \quad (2.5)$$

Thus

$$Z_n = \sum_{w \in \{X,Y\}^n} c(w)w. \quad (2.6)$$

For a prescribed word, the total number of cut patterns before zero terms are discarded is 2^{n-1} .

Proof. Write $U = e^X e^Y - 1$. Its nonempty-word coefficient is $a(w)$. A contribution to the coefficient of w in U^k is exactly a cut of w into k nonempty contiguous blocks; the contribution is the product of their coefficients. Substituting into the logarithm proves (2.5). Choosing a subset of the $n-1$ gaps is the same thing as choosing a cut pattern, proving the count. $\square$

For example, $c(XY) = 1 - 1/2 = 1/2$, while $c(YX) = -1/2$. For XXY , the one-block contribution is $1/2$, the two-block contribution is $-\frac{1}{2}(1 + \frac{1}{2})$, and the three-block contribution is $1/3$. Their sum is $1/12$. This small computation already explains why the coefficients are rational and why cancellation matters.

2.3 Any finite number of exponential factors

The same construction works without change for $\log(e^{A_1} \cdots e^{A_m})$. In the alphabet $A_1, \dots, A_m$, replace $a(w)$ by $(r_1! \cdots r_m!)^{-1}$ if $w = A_1^{r_1} \cdots A_m^{r_m}$, and by zero otherwise. If the factors are $e^{\alpha_j X}$ or $e^{\beta_j Y}$ with letters repeated in different positions, first use a distinct letter for each factor and then substitute. The resulting coefficients are finite polynomials in the scalar factor weights. This observation supplies all orders, not just leading errors, for the splitting formulas in [section 11](#).

3 Hopf algebra, primitive elements, and Dynkin's formula

3.1 The coproduct must take values in a completion

The relevant target of the coproduct on infinite series is

$$\widehat{T} \widehat{\otimes} \widehat{T} = \prod_{n \geq 0} \bigoplus_{p+q=n} V^{\otimes p} \otimes V^{\otimes q}.$$

The ordinary algebraic tensor product is generally too small. Define the continuous algebra homomorphism Δ by

$$\Delta X = X \otimes 1 + 1 \otimes X, \quad \Delta Y = Y \otimes 1 + 1 \otimes Y. \quad (3.1)$$

For a word $w = a_1 \cdots a_n$, let w_I be the subword at the positions $I \subseteq \{1, \dots, n\}$, in their original order. Then

$$\Delta w = \sum_{I \subseteq \{1, \dots, n\}} w_I \otimes w_{I^c}. \quad (3.2)$$

The empty subword is 1. Formula (3.2) follows by multiplying the n factors (3.1); it also proves coassociativity, because an iterated coproduct assigns each letter to one of three tensor factors independently of the order of assignment.

Let ε extract the constant term. Define the antipode on words by

$$S(a_1 \cdots a_n) = (-1)^n a_n \cdots a_1, \quad S(1) = 1. \quad (3.3)$$

It is an antiautomorphism. The convolution identities are

$$m(\text{id} \otimes S)\Delta w = m(S \otimes \text{id})\Delta w = \varepsilon(w)1. \quad (3.4)$$

Here m is multiplication. For completeness, the first identity can be proved inductively. If a is a letter and $\Delta v = \sum v_{(1)} \otimes v_{(2)}$, then the sum for av is

$$a \sum v_{(1)} S(v_{(2)}) - \sum v_{(1)} S(v_{(2)}) a,$$

which vanishes by the induction hypothesis. The degree-zero case is 1. The second identity follows in the same way, or by applying reversal. Thus $T(V)$, and continuously its completion, has the asserted Hopf algebra structure.

Definition 3.1. A positive-degree series P is *primitive* if $\Delta P = P \otimes 1 + 1 \otimes P$. A series G with constant term 1 is *grouplike* if $\Delta G = G \otimes G$.

The normalization for a grouplike series excludes the zero series, which would otherwise satisfy the displayed coproduct equation but would not belong to a group.

Lemma 3.2 (Exponentials of primitives). *e^P is grouplike if and only if P is primitive. Grouplike series form a group under multiplication.*

Proof. If P is primitive, the commuting elements $P \otimes 1$ and $1 \otimes P$ give

$$\Delta e^P = e^{\Delta P} = e^{P \otimes 1 + 1 \otimes P} = e^P \otimes e^P.$$

Conversely, if $G = e^P$ is grouplike, then $G \otimes G = (G \otimes 1)(1 \otimes G)$. These two factors commute, so

$$\Delta P = \log(\Delta G) = \log(G \otimes 1) + \log(1 \otimes G) = P \otimes 1 + 1 \otimes P.$$

Multiplicativity of Δ proves closure under products. A series with constant term 1 is invertible by its geometric series, and $\Delta(G^{-1}) = (G \otimes G)^{-1} = G^{-1} \otimes G^{-1}$. $\square$

3.2 An elementary projection proof of Friedrichs' theorem

Let N be the grading operator, $N(w) = |w|w$. Define

$$D = m(N \otimes S)\Delta. \quad (3.5)$$

This particular order of the two convolution factors is what produces *right*-nested rather than left-nested brackets.

Lemma 3.3 (Dynkin–Specht–Wever). *For every nonempty word w , $D(w) = [w]_R$. If P is a homogeneous primitive element of degree $n \geq 1$, then*

$$D(P) = nP. \quad (3.6)$$

Proof. For a letter a , the definition gives $D(a) = a$. Using the same coproduct expansion as above, and writing $|v_{(1)}|$ for degree, gives

$$\begin{aligned} D(av) &= \sum (|v_{(1)}| + 1)av_{(1)}S(v_{(2)}) - \sum |v_{(1)}|v_{(1)}S(v_{(2)})a \\ &= aD(v) - D(v)a + a\varepsilon(v). \end{aligned}$$

For nonempty v , this is $[a, D(v)]$, proving the first assertion by induction. If P is primitive, only its two primitive coproduct terms contribute, and $N(1) = 0$, so $D(P) = N(P)S(1) + N(1)S(P) = nP$. $\square$

Let $\mathcal{L} \subseteq T(V)$ be the Lie subalgebra generated by X, Y , and let $\mathcal{L}_n = \mathcal{L} \cap V^{\otimes n}$.

Theorem 3.4 (Friedrichs' characterization). *The primitive series are precisely $\prod_{n \geq 1} \mathcal{L}_n$, the formal degreewise sums of Lie polynomials.*

Proof. The generators are primitive. A commutator of primitives is primitive, since the mixed tensor factors commute and cancel in $[\Delta P, \Delta Q]$. Hence every Lie polynomial, and every convergent-in-degree sum of them, is primitive. Conversely, each homogeneous part P_n of a primitive series is primitive. By lemma 3.3,

$$P_n = \frac{1}{n} D(P_n) = \frac{1}{n} \sum_{|w|=n} [w] P_n [w]_{\text{R}} \in \mathcal{L}_n.$$

Characteristic zero is essential for division by n . □

Here and below $[w]P$ means the associative coefficient of the word w in P , not a commutator.

Corollary 3.5 (Formal BCH theorem). *Every $Z_n(X, Y)$ is a homogeneous Lie polynomial. In particular $Z - X - Y$ belongs degreewise to the Lie ideal generated by $[X, Y]$.*

Proof. X, Y are primitive, their exponentials are grouplike, the product is grouplike, and its logarithm is primitive. Apply theorem 3.4. A right-nested bracket of length at least two has innermost bracket either zero or $\pm[X, Y]$, proving the final statement. □

3.3 Dynkin's explicit all-orders formula

Applying D/n to the total-degree- n part of (2.3) proves the following formula, traditionally associated with Dynkin [2].

Theorem 3.6 (Dynkin formula). *In the completed free Lie algebra,*

$$Z(X, Y) = \sum_{k \geq 1} \frac{(-1)^{k-1}}{k} \sum_{\substack{r_i, s_i \geq 0 \\ r_i + s_i > 0 \ (1 \leq i \leq k)}} \frac{[X^{r_1} Y^{s_1} \dots X^{r_k} Y^{s_k}]_{\text{R}}}{(\sum_{i=1}^k (r_i + s_i)) \prod_{i=1}^k r_i! s_i!}. \quad (3.7)$$

In each total degree the sum is finite. A summand vanishes if $s_k > 1$, or if $s_k = 0$ and $r_k > 1$.

Proof. The projection argument proves the formula. The stated zero cases have innermost bracket $[Y, Y]$ or $[X, X]$ respectively. These are sufficient zero conditions, not a complete characterization of zero brackets. □

Combining this with the cut formula gives a particularly compact answer to the request for *all* terms:

$$Z_{p,q} = \frac{1}{p+q} \sum_{\substack{w \in \{X,Y\}^{p+q} \\ \#_X w = p, \ \#_Y w = q}} c(w) [w]_{\text{R}}, \quad p+q \geq 1, \quad (3.8)$$

where $c(w)$ is the finite sum (2.5). There are no unspecified coefficients or auxiliary recurrences in (3.8). The brackets form a spanning family rather than a basis, so different displayed terms can represent the same Lie element. This is redundancy, not ambiguity in Z .

3.4 Free Lie algebras and universal enveloping algebras

The Lie algebra $\mathcal{L}$ is free on X, Y , and its universal enveloping algebra is canonically $T(V)$. Here is a precise justification, including the point needed to substitute into an arbitrary Lie algebra.

The Poincaré–Birkhoff–Witt injectivity statement, proved in [section A](#), embeds any Lie algebra $\mathfrak{h}$ into $U(\mathfrak{h})$. Given $x, y \in \mathfrak{h}$, the associative homomorphism $T(V) \rightarrow U(\mathfrak{h})$ determined by $X \mapsto x, Y \mapsto y$ sends every iterated commutator into $\mathfrak{h}$. Its restriction therefore gives a Lie homomorphism $\mathcal{L} \rightarrow \mathfrak{h}$, unique because X, Y generate $\mathcal{L}$. This is exactly the universal property of the free Lie algebra.

The inclusion $\mathcal{L} \rightarrow T(V)$ extends to $U(\mathcal{L}) \rightarrow T(V)$, and the two generators give a homomorphism in the other direction. The two composites fix X, Y ; both associative algebras are generated by those elements, so the composites are identities. The Hopf structures agree on the generators, hence everywhere. This proves the enveloping-algebra assertion on the reference page.

For arbitrary $x, y \in \mathfrak{h}$ the safe universal expression is $\text{BCH}(tx, ty) \in \mathfrak{h}[[t]]$. An infinite sum in the uncompleted vector space $\mathfrak{h}$ has no meaning until a topology, a nilpotence hypothesis, or a convergence argument has been supplied.

4 Conjugation and differentiation of the exponential

4.1 Campbell’s identity and the full conjugation series

Theorem 4.1 (Campbell–Hadamard identity). *Formally, or for elements of a Banach algebra,*

$$e^{tX} Y e^{-tX} = e^{t \text{ad}_X} Y = \sum_{n \geq 0} \frac{t^n}{n!} \text{ad}_X^n Y. \quad (4.1)$$

Equivalently, $\text{Ad}_{e^X} = e^{\text{ad}_X}$.

Proof. Set $F(t) = e^{tX} Y e^{-tX}$. Direct differentiation gives $F'(t) = XF(t) - F(t)X = \text{ad}_X F(t)$ and $F(0) = Y$. The coefficient recursion uniquely solves this equation as the displayed series. In a Banach algebra, $\|\text{ad}_X^n Y\| \leq 2^n \|X\|^n \|Y\|$, so the series converges absolutely for every t and the same differentiation is legitimate. $\square$

An entirely associative version is

$$\text{ad}_X^n Y = \sum_{j=0}^n (-1)^j \binom{n}{j} X^{n-j} Y X^j. \quad (4.2)$$

Indeed, left and right multiplication by X commute, so their difference can be raised to the n th power using the ordinary binomial theorem. This proves every term of the Campbell expansion, including its iterated-commutator notation $[X, Y]_n$ on the reference page.

Conjugation is an algebra automorphism and therefore commutes with exponentiation. It follows that

$$e^X e^Y e^{-X} = \exp(e^X Y e^{-X}) = \exp(e^{\text{ad}_X} Y), \quad (4.3)$$

$$e^X e^Y = \exp \left(\sum_{n \geq 0} \frac{\text{ad}_X^n Y}{n!} \right) e^X. \quad (4.4)$$

These are the general braiding identities. Unlike a general BCH series, the adjoint exponential in (4.4) is an entire series for bounded operators.

4.2 The differential and its inverse

Theorem 4.2 (Differential of the exponential). *For an increment H ,*

$$(d \exp)_X(H) = \sum_{n \geq 1} \frac{1}{n!} \sum_{j=0}^{n-1} X^j H X^{n-1-j} = \int_0^1 e^{(1-s)X} H e^{sX} ds, \quad (4.5)$$

$$e^{-X} (d \exp)_X(H) = \phi_-(\text{ad}_X)H = \sum_{n \geq 0} \frac{(-1)^n}{(n+1)!} \text{ad}_X^n H, \quad (4.6)$$

$$(d \exp)_X(H) e^{-X} = \phi_+(\text{ad}_X)H = \sum_{n \geq 0} \frac{1}{(n+1)!} \text{ad}_X^n H. \quad (4.7)$$

All three series converge for arbitrary X, H in a Banach algebra.

Proof. Differentiating $(X + uH)^n$ at $u = 0$ inserts H in each possible position, giving the double sum. The coefficient of $X^p H X^q$ in the integral is

$$\frac{1}{p!q!} \int_0^1 (1-s)^p s^q ds = \frac{1}{(p+q+1)!}.$$

The integral identity follows; the elementary beta integral here follows by repeated integration by parts. Left multiplication by e^{-X} and [theorem 4.1](#) give (4.6). Right multiplication gives (4.7). The factorial denominators and the adjoint norm bound justify termwise operations in the analytic setting. $\square$

The same statements hold intrinsically for finite-dimensional Lie groups, where multiplication by e^{-X} means left or right translation of tangent vectors; an intrinsic derivation is supplied in [section 8](#).

4.3 Bernoulli coefficients, including the convention at degree one

Define the rational numbers b_n by

$$f(z) = \frac{z}{1 - e^{-z}} = \sum_{n \geq 0} b_n z^n = \sum_{n \geq 0} \frac{B_n^+}{n!} z^n. \quad (4.8)$$

Thus $B_1^+ = +1/2$, the convention used in the reference page's note. The alternative convention for $z/(e^z - 1)$ has $B_1^- = -1/2$ and $B_n^- = (-1)^n B_n^+$.

Proposition 4.3 (Complete Bernoulli coefficients). $b_0 = 1$, and for $n \geq 1$,

$$b_n = - \sum_{j=1}^n \frac{(-1)^j}{(j+1)!} b_{n-j}, \quad (4.9)$$

$$b_n = \sum_{k=1}^n (-1)^{n+k} \sum_{\substack{j_1 + \dots + j_k = n \\ j_i \geq 1}} \prod_{i=1}^k \frac{1}{(j_i + 1)!}. \quad (4.10)$$

Moreover,

$$f(z) = \frac{z}{2} \left(1 + \coth \frac{z}{2} \right) = 1 + \frac{z}{2} + \frac{z^2}{12} - \frac{z^4}{720} + \frac{z^6}{30240} - \frac{z^8}{1209600} + \dots \quad (4.11)$$

All odd coefficients beyond b_1 vanish, and the Taylor series of f has radius 2π .

Proof. The identity $\phi_-(z)f(z) = 1$ gives (4.9) by coefficient comparison. Expanding $(1 + \sum_{j \geq 1} (-1)^j z^j / (j+1)!)^{-1}$ as a geometric series gives (4.10). Algebraic simplification gives (4.11). The function $f(z) - z/2$ is even, proving the odd-coefficient assertion. The nonzero zeros of $1 - e^{-z}$ are $2\pi i\mathbb{Z} \setminus \{0\}$; the closest are genuine poles at $\pm 2\pi i$. The Taylor radius is therefore exactly 2π . $\square$

In particular, whenever inversion is legitimate,

$$H = f(\text{ad}_X)(e^{-X}(d \exp)_X(H)). \quad (4.12)$$

The quotient notation in this expression denotes its analytic value, not division by an invertible ad_X .

5 Differential equations, the integral formula, and all bidegrees

5.1 The differential equation for one moving factor

Theorem 5.1. *In the degree completion, or on a sufficiently small analytic neighborhood of $(0, 0)$, the function $Z(t) = \text{BCH}(X, tY)$ is characterized by*

$$Z'(t) = f(\text{ad}_{Z(t)})Y, \quad Z(0) = X. \quad (5.1)$$

Proof. For $G(t) = e^X e^{tY}$, its left logarithmic derivative is $G^{-1}G' = Y$. Substitute $G = e^{Z(t)}$ in (4.6): $Y = \phi_-(\text{ad}_Z)Z'$. Its formal inverse is $f(\text{ad}_Z)$, proving (5.1). Conversely, that equation implies that $e^{Z(t)}$ has the same left logarithmic derivative and initial value as $G(t)$. The equation $G' = GY$ has a unique solution, coefficient by coefficient formally, or by the ordinary uniqueness theorem analytically. $\square$

This is also a useful independent mechanism for constructing the BCH series: its right-hand side involves only iterated Lie brackets.

Theorem 5.2 (Poincaré integral formula). *With the same interpretations,*

$$\text{BCH}(X, Y) = X + \int_0^1 \psi(e^{\text{ad}_X} e^{t \text{ad}_Y}) Y dt, \quad (5.2)$$

where the all-orders expansion at $x = 1$ is

$$\psi(x) = 1 - \sum_{n \geq 1} \frac{(1-x)^n}{n(n+1)}. \quad (5.3)$$

In particular,

$$\psi(e^z) = f(z) = \sum_{n \geq 0} \frac{B_n^+}{n!} z^n. \quad (5.4)$$

Proof. Hadamard's identity and composition of conjugations give

$$e^{\text{ad}_{Z(t)}} = \text{Ad}(e^{Z(t)}) = \text{Ad}(e^X e^{tY}) = e^{\text{ad}_X} e^{t \text{ad}_Y}.$$

The formal logarithm of either side is $\text{ad}_{Z(t)}$; analytically this statement holds on the logarithm branch near the identity. Since $\psi(e^z) = z/(1 - e^{-z})$, equation (5.1) becomes the integrand of (5.2). Integrate from 0 to 1. For completeness, set $u = 1 - x$. Multiplying $\log(1 - u) = -\sum_{n \geq 1} u^n/n$ by $(1 - u)/(-u)$ gives (5.3), initially for $|u| < 1$, and also as a formal identity. Equation (5.4) then follows directly. In the formal integral, integration means integrating each polynomial coefficient in t , so no interchange of improper limits is involved. $\square$

There is no need for an invertible ad_Z in this argument. Likewise, $\psi(E)$ near $E = I$ is defined by its power series in $E - I$, not by multiplication by a nonexistent inverse of $E - I$.

5.2 Every power of the second variable

The notation $O(Y^2)$ hides a whole hierarchy which can be specified explicitly. Write

$$\text{BCH}(X, \varepsilon Y) = X + \sum_{m \geq 1} \varepsilon^m V_m(X, Y), \quad V_0 = X. \quad (5.5)$$

Each V_m is a formal series in X , homogeneous of degree m in Y .

Proposition 5.3 (All powers of Y). *With $b_k = B_k^+/k!$,*

$$mV_m = \sum_{k \geq 0} b_k \sum_{\substack{j_1 + \dots + j_k = m-1 \\ j_\ell \geq 0}} \text{ad}_{V_{j_1}} \cdots \text{ad}_{V_{j_k}} Y. \quad (5.6)$$

For $k = 0$ the inner expression is Y when $m = 1$, and is zero otherwise. In particular,

$$V_1 = f(\text{ad}_X)Y = \frac{\text{ad}_X}{2} \left(1 + \coth \frac{\text{ad}_X}{2} \right) Y, \quad (5.7)$$

$$V_2 = \frac{1}{2} \sum_{k \geq 1} b_k \sum_{\ell=0}^{k-1} \text{ad}_X^\ell \text{ad}_{V_1} \text{ad}_X^{k-1-\ell} Y. \quad (5.8)$$

Every V_m is thereby determined by $V_0, \dots, V_{m-1}$. A finite, nonrecursive alternative is

$$V_m = \sum_{p \geq 0} \frac{1}{p+m} \sum_{\substack{w \in \{X, Y\}^{p+m} \\ \#X(w)=p, \#Y(w)=m}} c(w)[w]_R, \quad (5.9)$$

where each $c(w)$ is the finite sum (2.5).

Proof. Expand the right-hand side of (5.1) in powers of ε . The coefficient of ε^{m-1} in the product of k adjoint maps is the inner sum in (5.6). Differentiation on the left contributes mV_m . At $m = 1$ every j_ℓ is zero; at $m = 2$ exactly one is one, giving (5.7) and (5.8). Finally, (5.9) is the bihomogeneous projection formula (3.8) summed over p . For a fixed total degree all sums are finite, including the apparently infinite sum over k in (5.6). $\square$

Here $O(Y^2)$ means membership in the closed subspace with at least two occurrences of Y , not an unspecified bound on $\|X\|$. The linear formula by itself is not an exact BCH formula unless a further relation annihilates all higher- Y terms.

5.3 An independent homogeneous recursion

Theorem 5.4. *Let $S = X + Y$, $D = X - Y$, and $Z(t) = \text{BCH}(tX, tY) = \sum_{n \geq 1} t^n Z_n$. Then*

$$Z' = S + \frac{1}{2}[D, Z] + \sum_{p \geq 1} b_{2p} \text{ad}_Z^{2p} S. \quad (5.10)$$

Consequently $Z_1 = S$ and, for $n \geq 2$,

$$nZ_n = \frac{1}{2}[D, Z_{n-1}] + \sum_{p=1}^{\lfloor (n-1)/2 \rfloor} b_{2p} \sum_{\substack{i_1 + \dots + i_{2p} = n-1 \\ i_j \geq 1}} \text{ad}_{Z_{i_1}} \cdots \text{ad}_{Z_{i_{2p}}} S. \quad (5.11)$$

This recursion uniquely constructs every homogeneous term by Lie operations, independently of the Hopf-algebra proof.

Proof. The right logarithmic derivative of $G(t) = e^{tX}e^{tY}$ is $X + e^{t\text{ad}_X}Y$. Also $e^{\text{ad}_Z}Y = e^{t\text{ad}_X}Y$, since $e^{t\text{ad}_Y}Y = Y$. The inverse of (4.7) is $z/(e^z - 1) = f(z) - z$. Hence

$$Z' = \frac{\text{ad}_Z}{e^{\text{ad}_Z} - 1}X + f(\text{ad}_Z)Y.$$

The two scalar functions have constant term 1, opposite linear terms $\mp z/2$, and identical even terms. Their sum applied to X, Y is exactly (5.10). Coefficient comparison gives (5.11). Its right-hand side uses only indices smaller than n , proving uniqueness and the constructive assertion. Reversing this reasoning shows that the resulting series exponentiates to $e^{tX}e^{tY}$; thus it coincides with the logarithmic construction. $\square$

6 All displayed low-degree terms and universal identities

6.1 The complete display through degree six

In the right-nested notation (1.2), the terms printed on the reference page are exactly

$$Z_1 = X + Y, \tag{6.1}$$

$$Z_2 = \frac{1}{2}[XY]_{\text{R}}, \tag{6.2}$$

$$Z_3 = \frac{1}{12}([XXY]_{\text{R}} + [YYX]_{\text{R}}), \tag{6.3}$$

$$Z_4 = -\frac{1}{24}[YXXY]_{\text{R}}, \tag{6.4}$$

$$Z_5 = -\frac{1}{720}([YYYYX]_{\text{R}} + [XXXXXY]_{\text{R}}) + \frac{1}{360}([XYYYX]_{\text{R}} + [YXXXY]_{\text{R}}) + \frac{1}{120}([YXYXY]_{\text{R}} + [XYXYX]_{\text{R}}), \tag{6.5}$$

$$Z_6 = \frac{1}{240}[YXYXY]_{\text{R}} + \frac{1}{720}([XYXXXY]_{\text{R}} - [XXYYXY]_{\text{R}}) + \frac{1}{1440}([XYYYXY]_{\text{R}} - [XXYXXY]_{\text{R}}). \tag{6.6}$$

For example, $[XYXXXY]_{\text{R}} = [X, [Y, [X, [X, [X, Y]]]]$. These displays do not select a linearly independent basis of the free Lie algebra; Jacobi identities can produce quite different-looking but equal answers.

Proposition 6.1. *Equations (6.1)–(6.6) are the homogeneous BCH components of degrees one through six.*

Proof. We give both a derivation and finite coefficient certificates, so the higher displayed degrees need not be accepted on the authority of a computer program. Put $C = [X, Y]$. For $n = 2$, (5.11) gives $2Z_2 = \frac{1}{2}[D, S] = C$. For $n = 3$ it gives $3Z_3 = \frac{1}{4}[D, C]$, since $\text{ad}_S^2 S = 0$. Thus $Z_3 = \frac{1}{12}([X, C] - [Y, C])$, equal to (6.3). For $n = 4$ the same recurrence gives

$$4Z_4 = \frac{1}{24}[D, [D, C]] - \frac{1}{24}[S, [S, C]] = -\frac{1}{6}[Y, [X, C]].$$

The last equality uses $[X, [Y, C]] = [Y, [X, C]]$, whose difference is $[[X, Y], C] = [C, C] = 0$. This proves (6.4).

For degrees five and six, expand the right brackets in (6.5)–(6.6) using $[A, B] = AB - BA$. The coefficient of any word w in this expansion is compared with the explicit rational number (2.5). All 2^5 and 2^6 values of the latter, including zeros, are printed in section B; expanding the respective displays gives exactly those tables. This is a finite coefficientwise certificate of the two identities in the free associative algebra, hence in every associative specialization. Alternatively, substitute $Z_1, \dots, Z_4$ in (5.11) for $n = 5$, and then $Z_1, \dots, Z_5$ for $n = 6$; ordinary bracket expansion gives the same tables. The supplied exact-arithmetic implementation performs all three comparisons independently. $\square$

An equality of noncommutative polynomials is fully determined by these word coefficients. Thus a complete coefficient table, together with the finite formula that computes each entry, is a proof certificate, not a numerical approximation or a test on finitely many matrices.

6.2 The associative display through degree four

Expanding (6.1)–(6.4) gives the other low-order formula on the reference page:

$$\begin{aligned} Z_1 &= X + Y, \\ Z_2 &= \frac{1}{2}(XY - YX), \\ Z_3 &= \frac{1}{12}(X^2Y + XY^2 - 2XYX + Y^2X + YX^2 - 2YXY), \\ Z_4 &= \frac{1}{24}(X^2Y^2 - 2XYXY - Y^2X^2 + 2YXYX). \end{aligned} \tag{6.7}$$

For example, $[Y, [X, [X, Y]]] = 2XYXY - 2YXYX + Y^2X^2 - X^2Y^2$, which proves the last line directly.

6.3 Symmetry, associativity, and functoriality

Proposition 6.2. *As universal formal series,*

$$\text{BCH}(Y, X) = -\text{BCH}(-X, -Y), \tag{6.8}$$

$$Z_n(Y, X) = (-1)^{n+1}Z_n(X, Y), \tag{6.9}$$

$$\text{BCH}(\text{BCH}(X, Y), T) = \text{BCH}(X, \text{BCH}(Y, T)). \tag{6.10}$$

Moreover, $\text{BCH}(X, 0) = \text{BCH}(0, X) = X$ and $\text{BCH}(X, -X) = 0$. Every Lie-algebra homomorphism commutes with these series wherever formal substitution or analytic convergence is valid.

Proof. Taking inverses of $e^{-X}e^{-Y} = e^{\text{BCH}(-X, -Y)}$ gives $e^Ye^X = e^{-\text{BCH}(-X, -Y)}$. Formal uniqueness of the logarithm gives (6.8), and homogeneous comparison gives (6.9). Exponentiating both sides of (6.10) gives $e^Xe^Ye^T$, so the same uniqueness proves associativity. The identity and inverse assertions follow in the same way. A Lie homomorphism preserves every iterated bracket, so it preserves Dynkin’s formula term by term. In a convergent analytic setting a continuous linear homomorphism also preserves its limit; finite-dimensional homomorphisms are automatically continuous. $\square$

Let ρ reverse words: $\rho(a_1 \cdots a_n) = a_n \cdots a_1$. It is an anti-automorphism and $\rho(e^Xe^Y) = e^Ye^X$. Therefore

$$c(\rho(w)) = (-1)^{|w|+1}c(w). \tag{6.11}$$

Both swap symmetry and reversal symmetry provide useful independent checks on large coefficient tables.

7 Convergence, remainder estimates, and logarithm obstructions

The formal theorem is unconditional, but its numerical evaluation is not. This section proves the sufficient estimates stated on the reference page, as well as several sharper distinctions needed to interpret them. We make no claim that these elementary bounds are optimal; convergence and continuation have a considerably richer structure [4].

7.1 Absolute convergence in a Banach algebra

Let $\mathcal{A}$ be a real or complex unital Banach algebra with a submultiplicative norm. Write $a = \|X\| + \|Y\|$. The arguments below apply equally to a finite-dimensional matrix algebra. They use submultiplicativity, not a special choice of operator norm. In particular the Hilbert–Schmidt matrix norm is admissible: by Cauchy–Schwarz,

$$\sum_{i,j} |(AB)_{ij}|^2 \leq \sum_{i,j} \left(\sum_k |A_{ik}|^2 \right) \left(\sum_k |B_{kj}|^2 \right) = \|A\|_{\text{HS}}^2 \|B\|_{\text{HS}}^2.$$

The possible value $\|I\| > 1$ in this norm causes no difficulty, because all the majorized logarithmic terms have positive degree.

Theorem 7.1 (Two explicit convergence domains). *If $a < \log 2$, the expanded associative series (2.3) is absolutely convergent, the homogeneous BCH series converges absolutely, and its sum Z satisfies $e^Z = e^X e^Y$. More precisely,*

$$\sum_{n \geq 1} \|Z_n(X, Y)\| \leq H(a), \quad H(u) = -\log(2 - e^u). \quad (7.1)$$

The unreduced multiple sum of nested commutators in Dynkin’s formula is absolutely convergent whenever $a < \frac{1}{2} \log 2$, with bound

$$\sum_{\text{all Dynkin summands}} \|\text{summand}\| \leq -\frac{1}{2} \log(2 - e^{2a}). \quad (7.2)$$

Proof. First,

$$\|e^X e^Y - I\| \leq \sum_{r+s > 0} \frac{\|X\|^r \|Y\|^s}{r!s!} = e^a - 1.$$

The sum of norms of all expanded terms of the k th logarithmic power is at most $(e^a - 1)^k/k$. If $a < \log 2$, summing this geometric-logarithmic majorant gives $-\log(2 - e^a)$. Absolute convergence allows regrouping by degree, and establishes (7.1). The analytic identity $\exp \log(I + U) = I + U$ for $\|U\| < 1$ follows by absolutely convergent scalar-series composition, exactly as in the formal proof. Thus the constructed Z exponentiates to $e^X e^Y$.

For the Dynkin sum, a right bracket of length N has norm at most 2^{N-1} times the product of its letter norms, by iterating $\|[A, B]\| \leq 2\|A\|\|B\|$. In (3.7) discard the factor $1/N \leq 1$ and majorize the remaining degree- N terms using 2^{N-1} . The k -block sum is then at most $(e^{2a} - 1)^k/(2k)$. Summing proves (7.2). $\square$

The two radii concern different ways of summing expressions. The larger bound applies to the series of already-collected homogeneous Lie polynomials, whereas the elementary smaller bound controls all the unreduced terms in Dynkin’s multiple sum separately.

The additional condition $\|Z\| < \log 2$ mentioned in the reference page is not required for the logarithm *constructed* by the Mercator series. Such an additional smallness condition can be useful when applying $\log(e^Z) = Z$ to an independently proposed logarithm, because that reverse identity requires a branch choice. Our proof constructs the branch and never assumes a bound on an unknown Z .

Corollary 7.2 (Explicit truncation error). *If $r > 1$ and $ra < \log 2$, then*

$$\left\| Z - \sum_{n=1}^N Z_n \right\| \leq r^{-N-1} [-\log(2 - e^{ra})]. \quad (7.3)$$

For a Banach Lie algebra with $\|[u, v]\| \leq K\|u\|\|v\|$, $K > 0$, Dynkin’s sum is absolutely convergent if $Ka < \log 2$, with total norm bounded by $K^{-1}[-\log(2 - e^{Ka})]$.

Proof. Homogeneity and (7.1) give $\sum_{n \geq 1} r^n \|Z_n\| \leq H(ra)$. Every term with $n > N$ therefore contributes at most r^{-N-1} times its weighted norm. For the Lie-algebra assertion replace the factor 2^{N-1} in the proof of (7.2) by K^{N-1} . If the bracket is zero, BCH simply terminates at $X + Y$ and no positive K is needed. $\square$

Every finite-dimensional Lie algebra admits such a K , because a bilinear map is bounded on the product of unit spheres. This supplies an elementary local convergence proof without requiring a matrix representation of the Lie algebra.

7.2 The trace identity and its branch restriction

Proposition 7.3. *For finite matrices on the local BCH branch,*

$$\mathrm{tr} \log(e^X e^Y) = \mathrm{tr} X + \mathrm{tr} Y. \quad (7.4)$$

For an arbitrary matrix logarithm L of $e^X e^Y$ over $\mathbb{C}$, the unconditional assertion is only

$$\mathrm{tr} L - \mathrm{tr} X - \mathrm{tr} Y \in 2\pi i \mathbb{Z}. \quad (7.5)$$

Proof. The equality $\mathrm{tr}(AB) = \mathrm{tr}(BA)$ follows by interchanging the two finite summation indices. Therefore every commutator has zero trace, and all Z_n for $n \geq 2$ have zero trace. Continuity of trace proves (7.4) wherever the BCH series converges. To prove the second statement, triangularize a complex matrix A : an eigenvector exists, and induction on the quotient dimension gives an upper triangular matrix with diagonal $\lambda_1, \dots, \lambda_d$. Its exponential is upper triangular with diagonal $e^{\lambda_1}, \dots, e^{\lambda_d}$. Hence $\det e^A = e^{\mathrm{tr} A}$. Apply this identity to L, X, Y and use multiplicativity of determinant. The scalar identity $e^z = 1$ holds exactly when $z \in 2\pi i \mathbb{Z}$, giving (7.5). $\square$

For example, take scalar $X = Y = 3\pi i/4$. Their sum is $3\pi i/2$, but the principal logarithm of $e^X e^Y = -i$ is $-\pi i/2$. Thus the word “logarithm” cannot be left branch-independent in (7.4).

7.3 A complete proof of the 2×2 obstruction

Theorem 7.4. *For*

$$X = \begin{pmatrix} 0 & i\pi \\ i\pi & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad (7.6)$$

the product is

$$e^X e^Y = \begin{pmatrix} -1 & -1 \\ 0 & -1 \end{pmatrix}. \quad (7.7)$$

It has no traceless complex logarithm. Its homogeneous BCH series therefore diverges, although the product does have logarithms in $\mathfrak{gl}_2(\mathbb{C})$.

Proof. Since $X^2 = -\pi^2 I$, the even and odd exponential series give $e^X = -I$. Since $Y^2 = 0$, $e^Y = I + Y$, proving (7.7). Any logarithm L commutes with its own exponential, hence with Y . Writing out $LY = YL$ shows that the centralizer of Y consists of matrices $aI + bY$. For such a matrix, $e^{aI+bY} = e^a(I + bY)$. Equality with $-I - Y$ forces $e^a = -1$ and $b = 1$. Thus all logarithms are

$$L = (2k+1)\pi i I + Y, \quad k \in \mathbb{Z}, \quad (7.8)$$

and each has nonzero trace $2(2k+1)\pi i$.

It remains to justify that even conditional convergence of BCH would contradict this fact. Suppose $\sum_{n \geq 1} Z_n(X, Y) = S$ converged. Its matrix power series $Z(t) = \sum_{n \geq 1} t^n Z_n$ would converge for $|t| < 1$, and $e^{Z(t)} = e^{tX} e^{tY}$ there: the identity holds near zero by the local theorem and then throughout the disk by analytic uniqueness, entry by entry. Abel's limit theorem gives $Z(t) \rightarrow S$ as real $t \uparrow 1$. To see that theorem directly, let $S_m = \sum_{n=1}^m Z_n$. Summation by parts gives $Z(t) = (1-t) \sum_{m \geq 1} t^m S_m$; split the sum into a fixed finite head and a tail where S_m is uniformly close to S . The head tends to zero and the tail tends to S . Continuity of the exponential now yields $e^S = e^X e^Y$. But every Z_n is traceless, so S is traceless, contradicting (7.8). $\square$

This example simultaneously distinguishes existence of a logarithm, existence of a logarithm in the specified Lie algebra, and convergence of the BCH series. They are genuinely different questions.

8 Lie groups, local multiplication, and integration

8.1 Matrix groups and the intrinsic differential

For a matrix Lie group G , its Lie algebra is the tangent space at the identity. The left-invariant field associated with a tangent matrix X is $X^L(g) = gX$. Its bracket with Y^L is $g(XY - YX)$, as follows by differentiating the two vector fields. Thus the induced Lie bracket is the matrix commutator. The integral curve through I of X^L solves $g' = gX$, $g(0) = I$, and its solution is the ordinary matrix exponential e^{tX} .

The relevant assertions do not require a matrix embedding. For an arbitrary finite-dimensional Lie group define its left Maurer–Cartan form by $\theta_g = (L_{g^{-1}})_*$. On left-invariant fields it takes constant values. The formula for the exterior derivative of a one-form consequently gives

$$d\theta + \frac{1}{2}[\theta, \theta] = 0. \quad (8.1)$$

Indeed, evaluated on U^L, V^L , the first term is $-[U, V]$ and the second is $[U, V]$; left-invariant fields span each tangent space.

For an intrinsic proof of (4.6), consider $g(t, u) = \exp(t(X + uH))$ and set $\alpha = \theta(\partial_t g)$, $\beta = \theta(\partial_u g)$. Here $\alpha = X + uH$. Pulling back (8.1) gives $\partial_t \beta - \partial_u \alpha = -[\alpha, \beta]$. At $u = 0$, therefore,

$$\partial_t \beta = H - \text{ad}_X \beta, \quad \beta(0) = 0.$$

Solving this linear equation gives $\beta(1) = \int_0^1 e^{-s \text{ad}_X} H ds = \phi_-(\text{ad}_X)H$. This is exactly the left logarithmic differential of the exponential. In particular $(d \exp)_0 = \text{id}$.

Theorem 8.1 (Local BCH multiplication). *The exponential map of a finite-dimensional real or complex Lie group is a local diffeomorphism at zero. In its local inverse chart,*

$$\exp X \exp Y = \exp \text{BCH}(X, Y) \quad (8.2)$$

for sufficiently small X, Y . Hence the local multiplication law is determined entirely by the Lie bracket.

Proof. The inverse function theorem applies because $(d \exp)_0 = \text{id}$. Choose a norm and a bracket bound K as in corollary 7.2; the universal BCH series then converges in a small ball in the Lie algebra. Its termwise differential equation (5.1) is valid in a smaller ball, by uniform convergence on compact subballs, or directly by the convergent majorant with a slightly larger scaling parameter. The intrinsic differential just proved shows that $\exp Z(t)$ has

left logarithmic derivative Y and starts at $\exp X$. Uniqueness for the corresponding smooth ordinary differential equation gives $\exp Z(t) = \exp X \exp(tY)$. At $t = 1$ this is (8.2). Local injectivity of the exponential selects this logarithm uniquely. $\square$

The local assertion is not a global classification: the additive Lie group $\mathbb{R}$ and the circle $\{e^{it} : t \in \mathbb{R}\}$ have the same one-dimensional abelian Lie algebra, but one is noncompact and the other compact, so they cannot be isomorphic as Lie groups.

8.2 Homomorphisms and representations

Theorem 8.2 (Local Lie correspondence). *A Lie-algebra homomorphism $A : \mathfrak{g} \rightarrow \mathfrak{h}$ defines a unique germ at the identity of a Lie-group homomorphism by*

$$F(\exp_G X) = \exp_H(AX). \quad (8.3)$$

Conversely, the differential at the identity of a smooth local group homomorphism preserves Lie brackets.

Proof. On exponential neighborhoods formula (8.3) defines a smooth map. Naturality of BCH and theorem 8.1 show that it preserves products whenever the inputs and their product remain in the chosen neighborhoods. For the converse, differentiating $F(gh) = F(g)F(h)$ shows that F relates the left-invariant field of X to that of $(dF)_e X$. The chain rule for vector-field brackets then gives $(dF)_e[X, Y] = [(dF)_e X, (dF)_e Y]$. Finally, a local homomorphism sends the integral curve $\exp_G(tX)$ to $\exp_H(t(dF)_e X)$, proving uniqueness of its germ. $\square$

Theorem 8.3 (Integration from a simply connected group). *If G is connected and simply connected and H is a Lie group, every Lie-algebra homomorphism $A : \mathfrak{g} \rightarrow \mathfrak{h}$ integrates to a unique smooth homomorphism $F : G \rightarrow H$.*

Proof. Start with the local homomorphism F_0 of theorem 8.2, and shrink its identity neighborhood so that sufficiently short products are covered by the local multiplication rule. Given a path γ from e to g , choose a partition fine enough that all increments $\gamma(t_j)^{-1}\gamma(t_{j+1})$ lie in this neighborhood, and set

$$F_\gamma(g) = \prod_j F_0(\gamma(t_j)^{-1}\gamma(t_{j+1})).$$

Inserting a partition point does not change the product, because the old increment is the product of the two new increments and F_0 preserves that small product. A common refinement therefore shows independence of a sufficiently fine partition.

For homotopy invariance, cover the compact parameter square of an endpoint-fixed homotopy by small rectangles on which all group values differ by elements of a still smaller identity neighborhood. Uniform continuity provides a finite rectangular subdivision with this property. Around each small rectangle, the four increments multiply to the identity. By further shrinking at the outset, all partial products around such a rectangle are in the domain where F_0 is multiplicative. Their images therefore multiply to the identity as well. Replacing one edge path of a rectangle by the other preserves its product. Sweeping across the grid proves equality for its two boundary paths; the endpoint edges contribute identities. The partition argument handles the actual boundary curves rather than just their grid descriptions. Thus F_γ depends only on the endpoint-fixed homotopy class of γ .

Simple connectedness makes it depend only on g ; call it $F(g)$. Concatenate a path to g with the left translate by g of a path to h . The increments of the second path are the

original increments of the path to h , so $F(gh) = F(g)F(h)$. For u close to e , appending a short path from g to gu gives $F(gu) = F(g)F_0(u)$, proving smoothness. Any other homomorphism with differential A agrees with F_0 near the identity and hence on finite products of such a neighborhood. These products form an open subgroup whose cosets are open; connectedness makes it all of G . This proves uniqueness. $\square$

Taking $H = \text{GL}(W)$ proves the corresponding integration statement for finite-dimensional representations. Without simple connectedness, the path construction can have a nontrivial value around a loop; this is the precise additional global obstruction absent from the local BCH law.

8.3 Nilpotent algebras and a global polynomial group law

Proposition 8.4. *If a Lie algebra is nilpotent of class at most c , every BCH term of degree greater than c vanishes. On its underlying vector space,*

$$x * y = \sum_{n=1}^c Z_n(x, y) \quad (8.4)$$

defines a Lie group with identity 0, inverse $-x$, original Lie bracket, and exponential map equal to the identity of the vector space. This group is connected and simply connected.

Proof. For the lower central series $\mathfrak{g}_1 = \mathfrak{g}$, $\mathfrak{g}_{j+1} = [\mathfrak{g}, \mathfrak{g}_j]$, induction using Jacobi shows $[\mathfrak{g}_p, \mathfrak{g}_q] \subseteq \mathfrak{g}_{p+q}$. Every Lie monomial of length n therefore lies in $\mathfrak{g}_n$ and vanishes for $n > c$. The formal associativity, identity, and inverse identities of [proposition 6.2](#) now become finite polynomial identities on all of $\mathfrak{g}$. Thus (8.4) is a smooth global group law. Its quadratic term is $[x, y]/2$; differentiating the group commutator $(tx) * (sy) * (-tx) * (-sy)$ at $t = s = 0$ gives $[x, y]$. Alternatively this follows by direct differentiation of its left-invariant fields. Since all brackets of multiples of a fixed x vanish, $(tx) * (sx) = (t + s)x$. These are exactly the one-parameter groups, so the exponential is $x \mapsto x$. The underlying manifold is a vector space and is therefore connected and simply connected. $\square$

If a given connected simply connected Lie group has this nilpotent Lie algebra, integrating the identity Lie-algebra maps in both directions by [theorem 8.3](#) gives inverse group isomorphisms. Its exponential is consequently a global diffeomorphism. For any other connected group with the same Lie algebra, integration from the polynomial group still shows that its exponentials satisfy the same finite BCH multiplication formula.

9 Exact special cases and their analytic limits

9.1 Commuting inputs and a central commutator

Theorem 9.1 (Central-commutator BCH). *Suppose $C = [X, Y]$ satisfies $[X, C] = [Y, C] = 0$. Then*

$$e^X e^Y = e^{X+Y+C/2}, \quad (9.1)$$

$$e^{X+Y} = e^X e^Y e^{-C/2} = e^{X/2} e^Y e^{X/2}, \quad (9.2)$$

$$e^X e^Y e^{-X} e^{-Y} = e^C. \quad (9.3)$$

These are formal identities, global identities for bounded elements of a Banach algebra, and global identities for elements of a finite-dimensional Lie group satisfying these bracket relations. In particular, $[X, Y] = 0$ gives $e^X e^Y = e^{X+Y}$.

Proof. Hadamard's identity truncates to $e^{sX}Ye^{-sX} = Y + sC$. Thus the right logarithmic derivative of $G(s) = e^{sX}e^{sY}$ is $X + Y + sC$. The function $E(s) = e^{s(X+Y)+s^2C/2}$ has the same derivative, since $X + Y$ and C commute. Both start at the identity, proving (9.1) at $s = 1$ by uniqueness. The intrinsic logarithmic derivative gives the identical proof in a Lie group. Moving the central exponential to the other side gives the first part of (9.2). Applying (9.1) twice to $X/2, Y, X/2$ cancels the two half-commutator corrections and gives its symmetric part. Finally, conjugation sends Y to $Y + C$, so $e^Xe^Ye^{-X}e^{-Y} = e^{Y+C}e^{-Y} = e^C$. $\square$

This proof is global and does not infer convergence from the formal truncation alone. For unbounded operators a common invariant domain and integrability assumptions are still necessary; see [section 13](#).

If all pairwise commutators of $A_1, \dots, A_m$ commute with every A_j , induction on m gives the full finite-factor identity

$$\prod_{j=1}^m e^{A_j} = \exp \left(\sum_{j=1}^m A_j + \frac{1}{2} \sum_{j < k} [A_j, A_k] \right). \quad (9.4)$$

Indeed, the commutator of the exponent for the first $m-1$ factors with A_m is just $\sum_{j < m} [A_j, A_m]$, since its central part commutes with A_m . The two-factor formula completes the induction. For a basis P, Q, C with $[P, Q] = C$ and all other brackets zero, (8.4) is the Heisenberg law

$$(x, y, z) * (x', y', z') = (x + x', y + y', z + z' + \frac{1}{2}(xy' - yx')). \quad (9.5)$$

This also proves directly that the Heisenberg algebra is two-step nilpotent and that all higher BCH terms vanish.

9.2 An adjoint eigenvector and exact resummation

Theorem 9.2. *Suppose $[X, Y] = sY$, where s is a scalar. Then*

$$e^Xe^Y = e^{e^sY}e^X, \quad (9.6)$$

$$e^{X+cY} = e^Xe^{c\phi_-(s)Y} \quad (c \text{ scalar}). \quad (9.7)$$

If $s \notin 2\pi i\mathbb{Z} \setminus \{0\}$, this implies

$$e^Xe^Y = \exp \left(X + \frac{s}{1 - e^{-s}} Y \right), \quad (9.8)$$

with coefficient 1 at $s = 0$. These exponential identities are global in the same bounded-algebra or finite-dimensional-group settings as [theorem 9.1](#). The local BCH Taylor series in this case is $X + f(s)Y$ with f expanded at zero; resummation and Taylor convergence are distinct assertions.

Proof. The relation implies $\text{ad}_X^n Y = s^n Y$, so Hadamard's formula gives $e^XYe^{-X} = e^sY$, proving (9.6) by conjugating the exponential. For (9.7), put

$$q(t) = c \int_0^t e^{-su} du, \quad G(t) = e^{tX}e^{q(t)Y}.$$

Its right logarithmic derivative is $X + q'(t)e^{ts}Y = X + cY$. Since $G(0) = I$, it equals $e^{t(X+cY)}$. At $t = 1$, $q(1) = c\phi_-(s)$, proving the factorization. Choosing $c = 1/\phi_-(s) = f(s)$ is possible exactly under the stated nonresonance condition and proves (9.8).

For the formal BCH assertion, the Lie algebra generated by X, Y is contained in their span, and its ideal spanned by Y is abelian. Every bracket monomial containing at least two copies of Y therefore vanishes: induct on a binary bracket expression, using that a bracket containing one Y is a multiple of Y , while an all- X bracket of length at least two is zero. Consequently only (5.7) remains. Applying $\text{ad}_X^n Y = s^n Y$ gives the Taylor expansion of $f(s)$. $\square$

At a nonzero resonance the restriction is genuine. Take

$$X = \begin{pmatrix} 2\pi i & 0 \\ 0 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Then $[X, Y] = 2\pi i Y$, $e^X e^Y = I + Y$, whereas (9.7) gives $e^{X+cY} = I$ for every c . There is nevertheless a logarithm of the product, namely Y . Thus the failure concerns the particular form $X + cY$, not the existence of any logarithm.

There is also a nonresonant distinction. If $Y \neq 0$ and $s \neq 0$,

$$\text{BCH}(tX, tY) = tX + tf(ts)Y \quad (9.9)$$

as a Taylor series at $t = 0$, and its radius is exactly $2\pi/|s|$, because the poles at $ts = \pm 2\pi i$ have nonzero residues in the Y direction. Thus real $s > 2\pi$ can make this series divergent at $t = 1$, although the resummed exponential identity (9.8) remains valid. The conjugation law (9.6), being an entire exponential identity, has no such resonance exclusion at all.

10 Zassenhaus factorization to every order

The complementary problem is to factor the exponential of a sum into exponentials of homogeneous Lie corrections. We first solve this in the degree completion, then prove a concrete analytic convergence theorem. Recursive constructions in this direction are also central in the computational literature [3].

10.1 Existence, uniqueness, and primitive corrections

Theorem 10.1 (Formal Zassenhaus product). *There is a unique sequence of homogeneous Lie polynomials C_n of degree n , $n \geq 2$, for which*

$$e^{t(X+Y)} = e^{tX} e^{tY} e^{t^2 C_2} e^{t^3 C_3} e^{t^4 C_4} \dots \quad (10.1)$$

The infinite product is read from left to right in increasing degree and is defined coefficientwise in t .

Proof. Set

$$R_1(t) = e^{-tY} e^{-tX} e^{t(X+Y)} = I + O(t^2). \quad (10.2)$$

Inductively, if $R_{n-1} = I + O(t^n)$, define

$$C_n = [t^n] R_{n-1}(t), \quad R_n(t) = e^{-t^n C_n} R_{n-1}(t). \quad (10.3)$$

Then $R_n = I + O(t^{n+1})$. Each residual is group-like: this is true initially by lemma 3.2, and remains true after removing the exponential of a primitive coefficient. To verify the induction, compare degree n in $\Delta R_{n-1} = R_{n-1} \otimes R_{n-1}$. Its first nonconstant coefficient satisfies $\Delta C_n = C_n \otimes 1 + 1 \otimes C_n$. By theorem 3.4 it is a Lie polynomial. Its degree is n , since every input has matching degree in t and the letters.

Undoing the residuals gives $R_1 = e^{t^2 C_2} \dots e^{t^n C_n} R_n$. As $n \rightarrow \infty$, $R_n \rightarrow I$ coefficientwise, proving (10.1). In any proposed factorization, the first unmatched coefficient must be $[t^n] R_{n-1}$, proving uniqueness. $\square$

In the alternative display $e^{-tX}e^{t(X+Y)} = \prod_{n \geq 1} e^{t^n C_n}$, the convention *must* be $C_1 = Y$. The notation would otherwise omit a necessary first-degree term.

10.2 The displayed exponents and a differential algorithm

The first corrections are

$$C_2 = -\frac{1}{2}[X, Y], \quad (10.4)$$

$$C_3 = \frac{1}{6}[X, [X, Y]] + \frac{1}{3}[Y, [X, Y]], \quad (10.5)$$

$$\begin{aligned} C_4 &= -\frac{1}{24}([[[X, Y], X], X] + 3[[[X, Y], X], Y] + 3[[[X, Y], Y], Y]) \\ &= -\frac{1}{24}[X X X Y]_R - \frac{1}{8}[Y X X Y]_R - \frac{1}{8}[Y Y X Y]_R. \end{aligned} \quad (10.6)$$

Both versions of C_4 agree by antisymmetry; for example $[[[X, Y], X], Y] = [Y, [X, [X, Y]]]$.

For a complete algorithm using only brackets, set $F_n = R'_n R_n^{-1} = \sum_{k \geq 0} t^k f_{n,k}$. Direct differentiation of (10.2), using Hadamard's formula, yields

$$F_1 = e^{-t \operatorname{ad}_Y} (e^{-t \operatorname{ad}_X} - I)Y = \sum_{k \geq 1} (-t)^k \sum_{j=1}^k \frac{\operatorname{ad}_Y^{k-j} \operatorname{ad}_X^j Y}{(k-j)!j!}. \quad (10.7)$$

Indeed the right logarithmic derivatives of the three factors give $-Y - e^{-t \operatorname{ad}_Y} X + e^{-t \operatorname{ad}_Y} e^{-t \operatorname{ad}_X} (X + Y)$; the X terms cancel, and $e^{-t \operatorname{ad}_Y} Y = Y$. Differentiating (10.3) gives, for $n \geq 2$,

$$F_n = e^{-t^n \operatorname{ad}_{C_n}} F_{n-1} - n t^{n-1} C_n, \quad (10.8)$$

$$C_n = \frac{1}{n} f_{n-1, n-1}, \quad (10.9)$$

$$f_{n,k} = \sum_{j=0}^{\lfloor k/n \rfloor} \frac{(-1)^j}{j!} \operatorname{ad}_{C_n}^j f_{n-1, k-nj} - n C_n \delta_{k, n-1}. \quad (10.10)$$

Equation (10.9) follows from $R_{n-1} = I + t^n C_n + O(t^{n+1})$. These formulas are valid at every order and constitute a second construction of the same unique factorization.

For an explicit derivation of (10.4)–(10.6), write $C = [X, Y]$. Formula (10.7) begins

$$F_1 = -tC + t^2 \left(\frac{1}{2}[X, C] + [Y, C] \right) - t^3 \left(\frac{1}{6}[X, [X, C]] + \frac{1}{2}[Y, [X, C]] + \frac{1}{2}[Y, [Y, C]] \right) + O(t^4).$$

Dividing the t coefficient by 2 gives C_2 . Passing to F_2 removes that coefficient; its possible degree-three modification is $-[C_2, -C] = 0$. Hence its degree-two coefficient divided by 3 gives C_3 , and its degree-three coefficient divided by 4 gives C_4 . Removing C_3 does not affect the latter coefficient. This proves every displayed exponent without leaving implicit a higher-order approximation that could change it.

10.3 A finite nonrecursive formula using weighted trees

The previous algorithms determine all coefficients recursively. For a genuinely finite nonrecursive expression at every fixed order, first define the explicit associative polynomials

$$H_n = [t^n] R_1(t) = \sum_{a+b+c=n} \frac{(-1)^{a+b}}{a!b!c!} Y^a X^b (X+Y)^c. \quad (10.11)$$

Here $H_0 = I$ and $H_1 = 0$; each $(X + Y)^c$ is the sum of all 2^c words of length c . Coefficient extraction from the ordered product for R_1 gives

$$C_n = H_n - \sum_{\substack{k_2, \dots, k_{n-1} \geq 0 \\ \sum_{j=2}^{n-1} jk_j = n}} \frac{C_2^{k_2} \cdots C_{n-1}^{k_{n-1}}}{k_2! \cdots k_{n-1}!}. \quad (10.12)$$

The sum is empty for $n = 2, 3$. Every product in it has at least two factors. Importantly, its order is increasing in the degree of its factors: these polynomials need not commute.

Definition 10.2. Let $\mathcal{T}_n$ consist of finite plane rooted trees with an integer weight at each vertex. The root has weight n , every weight is at least 2, and each internal vertex has at least two children. The child weights are weakly increasing from left to right and sum to the parent's weight. Leaves have no further restriction. For an internal vertex v , let $m_v(j)$ be the number of its children of weight j . Define

$$\omega(T) = (-1)^{\#\{\text{internal vertices}\}} \prod_{v \text{ internal}} \prod_{j \geq 2} \frac{1}{m_v(j)!}, \quad H(T) = \prod_{\ell \text{ leaf, left to right}} H_{\text{wt}(\ell)}. \quad (10.13)$$

Subtrees of equal root weight are still ordered; interchanging unequal such subtrees can change $H(T)$.

Theorem 10.3 (All-terms Zassenhaus tree formula). *For every $n \geq 2$,*

$$\boxed{C_n = \sum_{T \in \mathcal{T}_n} \omega(T) H(T)}. \quad (10.14)$$

This is a finite formula involving only factorials, sums, and products of X, Y . An explicitly nested-commutator version is

$$C_n = \frac{1}{n} \sum_{T \in \mathcal{T}_n} \omega(T) \sum_{w \in \{X, Y\}^n} ([w]H(T))[w]_{\text{R}}. \quad (10.15)$$

Proof. All leaf weights sum to n , so there are at most $\lfloor n/2 \rfloor$ leaves. With at least two children at every internal vertex there are fewer internal vertices than leaves. Thus only finitely many tree shapes and weight assignments occur. The tree consisting only of its root contributes H_n . For any other tree, let the root's child weights be the nondecreasing list $i_1, \dots, i_k$. Removing the root leaves an independently chosen tree in each $\mathcal{T}_{i_j}$. Its root contributes the negative sign and the factor $\prod_j m(j)!^{-1}$. Therefore the right-hand side of (10.14), grouped by root children, satisfies exactly (10.12), with initial values H_2, H_3 . Induction on n proves the formula. Since C_n is primitive, applying the right Dynkin projection D/n to it leaves it fixed. Applying that projection to its word expansion gives (10.15). $\square$

For orientation, the first nontrivial instances are

$$\begin{aligned} C_2 &= H_2, & C_3 &= H_3, & C_4 &= H_4 - \frac{1}{2}H_2^2, \\ C_5 &= H_5 - H_2H_3, \\ C_6 &= H_6 - H_2H_4 - \frac{1}{2}H_3^2 + \frac{1}{3}H_2^3. \end{aligned} \quad (10.16)$$

The last coefficient is $1/2 - 1/6 = 1/3$: one contribution comes from substituting $C_4 = H_4 - H_2^2/2$ in $-C_2C_4$, and the other from $-C_2^3/6$. This example illustrates why an unordered product would lose information at higher degrees.

10.4 A quantitative convergence theorem for the product

Theorem 10.4. *For bounded X, Y in a Banach algebra put $a = \|X\| + \|Y\|$. If $r > 0$ satisfies*

$$h(r) := e^{2ar} - 1 - 2ar < 2 \log 2 - 1, \quad (10.17)$$

then $\sum_{n \geq 2} r^n \|C_n\| < \infty$ and the Zassenhaus product converges in norm for $|t| \leq r$, to $e^{t(X+Y)}$. A particularly simple sufficient condition is $a|t| \leq 1/4$.

Proof. The three exponential factors defining H_n give $\|H_n\| \leq (2a)^n/n!$ for $n \geq 2$. Introduce a scalar formal series $c(z)$ with nonnegative coefficients by

$$c = h + (e^c - 1 - c), \quad h(z) = e^{2az} - 1 - 2az. \quad (10.18)$$

The coefficient of degree n in $e^c - 1 - c$ depends only on coefficients of c of degrees less than n , since c starts at degree 2. Thus this defines a unique nonnegative formal series. The triangle inequality in (10.12), and the fact that scalar factors commute, prove inductively that $\|C_n\| \leq [z^n]c(z)$: the scalar product coefficients are exactly those of $e^{c(z)}$ after removing the constant and linear terms in c .

To prove convergence rather than just formal majorization, choose $q \in (0, \log 2)$ with $h(r) = 2q + 1 - e^q$. Such a q exists under (10.17), since $2q + 1 - e^q$ is strictly increasing from 0 to $2 \log 2 - 1$ on that interval. The case $a = 0$ is trivial. Iterate $c^{(0)} = 0$ and $c^{(m+1)} = h + e^{c^{(m)}} - 1 - c^{(m)}$. Its coefficients increase monotonically and stabilize at each fixed degree to those of c . At $z = r$ the scalar values stay at most q , by induction and the monotonicity of $e^u - 1 - u$ for $u \geq 0$. Monotone convergence of nonnegative series gives $\sum_n [z^n]c r^n \leq q$.

It follows that $\sum \|t^n C_n\| \leq q$ for $|t| \leq r$. Successive products of exponentials are Cauchy in norm: a tail product differs from I by at most $\exp(\sum_{n > N} \|t^n C_n\|) - 1$, and the norm of the preceding product is uniformly bounded by the exponential of the total sum (up to the harmless norm of I). Absolute convergence permits coefficientwise multiplication, so the formal identity (10.1) proves equality of the analytic products for $|t| < r$, and continuity proves equality on the boundary. Finally $a|t| \leq 1/4$ implies $h(|t|) \leq e^{1/2} - 3/2 < 2 \log 2 - 1$, as is seen, for example, from the convergent scalar series or elementary numerical bounds $e^{1/2} < 1.649$ and $\log 2 > 0.693$. $\square$

This is an elementary sufficient criterion, not a claim about the maximal domain or the best constant in the literature.

If $[X, Y] = sY$, the factorization (9.7) at scale t gives a particularly simple all-orders answer:

$$C_n = \frac{(-1)^{n-1} s^{n-1}}{n!} Y \quad (n \geq 2). \quad (10.19)$$

Indeed, $t\phi_-(ts)Y = tY + \sum_{n \geq 2} (-1)^{n-1} t^n s^{n-1} Y/n!$, and all these exponents commute. This Zassenhaus product converges for every t , even in examples where the BCH Taylor series has finite radius. If the commutator is central instead, only $C_2 = -[X, Y]/2$ survives.

11 Trotter and Suzuki product formulas with full error series

The reference page mentions the Suzuki–Trotter decomposition as a consequence of Zassenhaus. Here we give the bounded-operator results and their all-order logarithmic expansions. They do not assert an unrestricted product theorem for unbounded semigroup generators. The classical sources treat broader settings [5, 6].

11.1 The ordinary product formula

Theorem 11.1. *For bounded X, Y and every fixed real or complex t ,*

$$\lim_{n \rightarrow \infty} (e^{tX/n} e^{tY/n})^n = e^{t(X+Y)} \quad (11.1)$$

in norm, locally uniformly in t . For sufficiently large n its complete logarithmic expansion is

$$(e^{tX/n} e^{tY/n})^n = \exp \left(t(X+Y) + \sum_{k \geq 2} t^k n^{1-k} Z_k(X, Y) \right). \quad (11.2)$$

The convergence error is $O(n^{-1})$, uniformly on each compact t -set.

Proof. When $|t|a/n < \log 2$, the one-step logarithm is the convergent BCH series. Since powers of a single exponential satisfy $(e^A)^n = e^{nA}$, multiplying its logarithm by n gives (11.2). The sum of terms starting at $k = 2$ is $O(n^{-1})$ uniformly for bounded t , by the convergence majorant on any fixed slightly larger disk in the variable t/n . For bounded A, B , differentiation of $e^{(1-s)A} e^{sB}$ gives the identity

$$e^B - e^A = \int_0^1 e^{(1-s)A} (B - A) e^{sB} ds.$$

Consequently $\|e^B - e^A\| \leq \|B - A\| e^{\max(\|A\|, \|B\|)}$ for a normalized norm; the same estimate with a fixed harmless multiplicative constant holds in the other admissible norms. Applying it to the exponents proves the stated limit and error. $\square$

Every coefficient of the error series is explicitly supplied by (2.5) or Dynkin's formula; it is not merely a formal symbol for an unknown remainder.

11.2 The symmetric product and arbitrary even order

Define

$$S_2(t) = e^{tX/2} e^{tY} e^{tX/2}. \quad (11.3)$$

Since $S_2(-t) = S_2(t)^{-1}$, its local logarithm is odd:

$$\log S_2(t) = t(X+Y) + \sum_{j \geq 1} t^{2j+1} E_{2j+1}. \quad (11.4)$$

A direct substitution into BCH through degree three gives

$$E_3 = -\frac{1}{24}[X, [X, Y]] - \frac{1}{12}[Y, [X, Y]]. \quad (11.5)$$

For clarity, one can obtain this by first writing $A = \text{BCH}(tX/2, tY) = tX/2 + tY + t^2[X, Y]/4 + t^3([X, [X, Y]]/48 - [Y, [X, Y]]/24) + O(t^4)$, and then expanding $\text{BCH}(A, tX/2)$ using (6.2)–(6.3). The quadratic terms cancel and the cubic terms combine to (11.5); all even terms vanish by the inverse symmetry. The multifactor version of theorem 2.2 gives every E_{2j+1} as a finite cut sum. It follows as above that $S_2(t/n)^n$ approximates $e^{t(X+Y)}$ with norm error $O(n^{-2})$ and has full logarithmic expansion $t(X+Y) + \sum_{j \geq 1} t^{2j+1} n^{-2j} E_{2j+1}$.

Theorem 11.2 (Suzuki's even-order recursion). *For $k \geq 2$, set*

$$p_k = \frac{1}{4 - 4^{1/(2k-1)}}, \quad (11.6)$$

and recursively define the symmetric product

$$S_{2k}(t) = S_{2k-2}(p_k t)^2 S_{2k-2}((1 - 4p_k)t) S_{2k-2}(p_k t)^2. \quad (11.7)$$

Then $\log S_{2k}(t) = t(X + Y) + O(t^{2k+1})$. For each fixed k , $S_{2k}(t/n)^n$ has error $O(n^{-2k})$ in norm, locally uniformly in t . Every term in its local logarithm has an explicit finite multifactor word-cut expansion.

Proof. Inductively suppose that S_{2k-2} is symmetric and its logarithm is $tS + t^{2k-1}E + O(t^{2k+1})$, $S = X + Y$. The palindromic product (11.7) is symmetric as well. Its linear terms add to tS because $4p_k + (1 - 4p_k) = 1$. Through degree $2k - 1$, all commutators between the linear terms vanish, since they are scalar multiples of S , and commutators involving E have degree at least $2k$. Thus the degree- $2k - 1$ error is

$$(4p_k^{2k-1} + (1 - 4p_k)^{2k-1})E = 0,$$

since $1 - 4p_k = -4^{1/(2k-1)}p_k$. The degree- $2k$ term also vanishes by symmetry, leaving an error of order $2k + 1$. This establishes the induction from (11.4). For fixed k the expression is a finite product of exponentials; its logarithm therefore converges near zero. Scaling by t/n and raising to the n th power yields a logarithmic error $O(n^{-2k})$, and the exponential estimate used in theorem 11.1 proves the norm assertion. Applying the multifactor coefficient formula to this finite product gives the all-orders expansion claimed. $\square$

The middle coefficient $1 - 4p_k$ is negative. Such negative steps are harmless for bounded operators but can be a serious obstruction for unbounded semigroup generators. The hypothesis in this section is therefore part of the result, not an optional technicality.

12 Maurer–Cartan coordinates, vielbeins, and all-order metrics

12.1 The infinitesimal formula and its full component expansion

Let $e_1, \dots, e_d$ be a basis of a finite-dimensional Lie algebra, with structure constants

$$[e_i, e_j] = f_{ij}^k e_k, \quad X = X^i e_i. \quad (12.1)$$

Repeated upper–lower indices are summed. For a differentiable Lie-algebra-valued function X , the differential formula already proved gives the complete infinitesimal identity

$$e^{-X} de^X = \sum_{n \geq 0} \frac{(-1)^n}{(n+1)!} \text{ad}_X^n(dX) = dX - \frac{1}{2}[X, dX] + \frac{1}{6}[X, [X, dX]] - \dots \quad (12.2)$$

In a nonmatrix Lie group, $e^{-X} de^X$ is the conventional notation for the pullback of the left Maurer–Cartan form, so the intrinsic proof in section 8 applies unchanged. The displayed component terms on the reference page are

$$\begin{aligned} e^{-X} de^X &= e_k dX^k - \frac{1}{2} X^i dX^j f_{ij}^k e_k + \frac{1}{6} X^i X^j dX^k f_{jk}^\ell f_{i\ell}^m e_m \\ &\quad + \text{terms with at least three factors } X^i. \end{aligned} \quad (12.3)$$

Each term follows by inserting (12.1) in the successive commutators; thus its indices and signs are fixed by the bracket convention rather than guessed from tensor notation.

Define the adjoint matrix and the left exponential-coordinate coframe matrix by

$$M_b^a = X^i f_{ib}^a, \quad W(X) = \phi_-(M) = \sum_{n \geq 0} \frac{(-1)^n}{(n+1)!} M^n. \quad (12.4)$$

Then every component at every order is specified by

$$e^{-X} de^X = e_a W_b^a(X) dX^b, \quad (12.5)$$

where for $n \geq 1$

$$(M^n)_b^a = \sum_{\substack{i_1, \dots, i_n \\ c_1, \dots, c_{n-1}}} X^{i_1} \dots X^{i_n} \prod_{r=1}^n f_{i_r c_r}^{c_{r-1}}, \quad c_0 = a, \quad c_n = b. \quad (12.6)$$

For $n = 1$ this formula is simply $X^i f_{ib}^a$, with the intermediate-index string absent. Formula (12.6) is ordinary matrix multiplication written out, and the factorial series in (12.4) converges for every finite matrix M .

Proposition 12.1 (Which matrix is the vielbein?). *The coframe coefficients of the Maurer–Cartan form in exponential coordinates are W_b^a , not M_b^a . When the Lie algebra has positive dimension, M is always singular, whereas $W(0) = I$ and W is invertible near zero. More precisely,*

$$W(X) \text{ is invertible} \iff \text{spec}(\text{ad}_X) \cap (2\pi i\mathbb{Z} \setminus \{0\}) = \emptyset, \quad (12.7)$$

where the spectrum is taken after complexification for a real Lie algebra. On this set $W^{-1} = f(M)$, interpreted by analytic functional calculus, not necessarily by a convergent Taylor series about zero at every such M .

Proof. Equation (12.5) identifies the coframe. If $X \neq 0$, then $MX = [X, X] = 0$; at $X = 0$, $M = 0$. Thus M cannot be an invertible frame matrix. By contrast the constant term of W is I . Triangularize M over $\mathbb{C}$. A power series in a triangular matrix has diagonal entries equal to that scalar function at the diagonal entries of M . Therefore $\det W = \prod_{\lambda \in \text{spec } M} \phi_-(\lambda)$, with eigenvalues counted with multiplicity. The zeros of ϕ_- are exactly the nonzero $2\pi i$ integers, proving (12.7). Where no zero is present, the analytic identity $\phi_- f = 1$ yields $W^{-1} = f(M)$. $\square$

Some authors reserve “vielbein” for the dual vector frame rather than its coframe. With that convention the frame coefficients are W^{-1} , still not M . The quotient $(I - e^{-M})M^{-1}$ used on the reference page is a shorthand for the entire function $\phi_-(M)$; a literal inverse of M does not exist in this setting. This is a correction of a mathematical misidentification on the page, not an alternative sign convention.

12.2 Pullbacks and the hypotheses needed for a metric

Choose a symmetric bilinear form B on the Lie algebra, with matrix $B_{ab} = B(e_a, e_b)$. Its left-invariant extension to the group evaluates tangent vectors by first applying the Maurer–Cartan form. For a smooth map $F : N \rightarrow G$, near any fixed q_0 a fixed left translation replaces $F(q)$ by $F(q_0)^{-1}F(q)$ without changing the pullback of a left-invariant tensor. We may therefore work in an exponential neighborhood and write $F(q) = \exp X(q)$. Set

$$E_\mu^a(q) = W_b^a(X(q)) \partial_\mu X^b(q). \quad (12.8)$$

The complete pullback formula is then

$$(F^*B)_{\mu\nu} = B_{ab} E_\mu^a E_\nu^b. \quad (12.9)$$

This follows immediately from the definition of pullback: evaluate B on $F_*\partial_\mu, F_*\partial_\nu$ and use (12.5). In exponential coordinates themselves, $X^b = q^b$, so the Jacobian is the identity and the page’s formula becomes

$$g_{ij} = B_{ab} W_i^a W_j^b, \quad g = W^T B W. \quad (12.10)$$

For an arbitrary map, the Jacobian factors in (12.8) cannot be omitted.

Several qualifications matter. A symmetric bilinear tensor need not be a metric: a pseudo-Riemannian metric must be nondegenerate, and a Riemannian metric must be positive definite. If B is positive definite and F is an immersion, (12.9) is positive definite. If B is indefinite, even an immersion can have degenerate pullback: the immersion $q \mapsto (q, q)$ into $\mathbb{R}^2$ with bilinear form $dx^2 - dy^2$ pulls it back to zero. A constant map pulls back every bilinear form to zero.

The Killing form is

$$B_K(u, v) = \text{tr}(\text{ad}_u \text{ad}_v), \quad (B_K)_{ab} = f_{ac}^d f_{bd}^c. \quad (12.11)$$

It is symmetric by cyclicity of trace. Its invariance is proved by

$$B_K([x, u], v) + B_K(u, [x, v]) = \text{tr}([\text{ad}_x, \text{ad}_u] \text{ad}_v + \text{ad}_u [\text{ad}_x, \text{ad}_v]) = 0,$$

using $\text{ad}_{[x, u]} = [\text{ad}_x, \text{ad}_u]$, which is Jacobi's identity. But the Killing form is not nondegenerate in every Lie algebra: in an abelian algebra every adjoint map is zero, so $B_K = 0$. Consequently calling its pullback a "metric" without additional nondegeneracy hypotheses is not a valid general assertion. The formula (12.10) is valid regardless of whether it defines a metric or only a possibly degenerate tensor.

12.3 All-orders metric coefficients and their even resummation

Without assuming invariance, substitution of (12.4) into (12.10) gives the absolutely convergent expansion

$$g_{ij} = \sum_{p, q \geq 0} \frac{(-1)^{p+q}}{(p+1)!(q+1)!} B_{ab} (M^p)^a_i (M^q)^b_j. \quad (12.12)$$

If B is invariant, the answer becomes considerably simpler.

Theorem 12.2 (Complete invariant-metric expansion). *For an invariant symmetric bilinear form B ,*

$$g = B \frac{e^M + e^{-M} - 2I}{M^2} = B \left(\frac{\sinh(M/2)}{M/2} \right)^2 \quad (12.13)$$

$$= B \sum_{n \geq 0} \frac{2M^{2n}}{(2n+2)!} = B \left(I + \frac{M^2}{12} + \frac{M^4}{360} + \frac{M^6}{20160} + \cdots \right). \quad (12.14)$$

All quotients in this display denote entire functions with their removable values at zero. No inverse of M or M^2 is assumed.

Proof. Invariance says $B([X, u], v) + B(u, [X, v]) = 0$, or $M^T B + B M = 0$. Hence $(M^p)^T B = (-1)^p B M^p$, by induction. The series for W therefore gives $W^T B = B \phi_-(-M)$. Multiply on the right by $W = \phi_-(M)$:

$$W^T B W = B \phi_-(-M) \phi_-(M) = B \frac{(e^M - I)(I - e^{-M})}{M^2}.$$

The numerator is $e^M + e^{-M} - 2I$. The scalar hyperbolic identity and its power series prove both displays. All operations are between convergent entire functions of the same matrix, so their factors commute as used in the simplification. The resulting matrix is symmetric because it was obtained as $W^T B W$. $\square$

If B and W are invertible, the inverse tensor is explicitly $g^{-1} = f(M) B^{-1} f(M)^T$. In a neighborhood of zero, expanding each $f(M)$ using (4.10) supplies all coefficients of this inverse as well.

12.4 A concrete three-dimensional resummation

Identify the rotation Lie algebra with $\mathbb{R}^3$ and bracket $[x, v] = x \times v$. Here $Mv = x \times v$ and $r = |x|$. The vector triple-product identity gives $M^2v = x(x \cdot v) - r^2v$ and hence $M^3 = -r^2M$. Summing the exponential series by reducing powers with this identity gives Rodrigues' formula

$$e^{-sM} = I - \frac{\sin(sr)}{r}M + \frac{1 - \cos(sr)}{r^2}M^2.$$

Integrating from $s = 0$ to 1 proves the complete coframe formula

$$W = I - \frac{1 - \cos r}{r^2}M + \frac{r - \sin r}{r^3}M^2, \quad (12.15)$$

with its Taylor values at $r = 0$. The ordinary dot product is invariant because $(x \times u) \cdot v + u \cdot (x \times v) = 0$. For $r > 0$, let $P_{\parallel} = xx^T/r^2$ and $P_{\perp} = I - P_{\parallel}$. On the parallel line $M = 0$; on the perpendicular plane $M^2 = -r^2I$. Formula (12.13) therefore gives

$$W^TW = P_{\parallel} + \left(\frac{\sin(r/2)}{r/2}\right)^2 P_{\perp}, \quad \det W = \left(\frac{\sin(r/2)}{r/2}\right)^2. \quad (12.16)$$

The determinant follows either from the eigenvalues $0, ir, -ir$ of M or directly from (12.15). Thus the coordinate coframe is singular at nonzero radii $2\pi, 4\pi, \dots$, precisely as (12.7) predicts. At zero the whole expression has the unambiguous limit I , despite the individual projector notation there being unnecessary.

13 Quantum-mechanical identities with rigorous interpretations

The algebraic calculations in the reference page are correct in the Heisenberg algebra. For operators on a Hilbert space, however, unboundedness and domains cannot be suppressed. We first prove the exponentiated identities in a concrete representation, then exhibit why a commutator identity on a dense domain alone is insufficient. This avoids using the Stone-von Neumann classification theorem, which is mentioned but not stated on the reference page.

13.1 Position, momentum, and the Weyl relations

On $L^2(\mathbb{R})$ fix $\hbar > 0$. On the Schwartz space $\mathcal{S}(\mathbb{R})$ let

$$(Qf)(x) = xf(x), \quad (Pf)(x) = -i\hbar f'(x). \quad (13.1)$$

Direct differentiation gives

$$[Q, P]f = i\hbar f \quad (f \in \mathcal{S}(\mathbb{R})). \quad (13.2)$$

The self-adjoint realizations are the usual multiplication operator and the Fourier-transform conjugate multiplication operator. Schwartz space is a core for these operators: cutoff and smoothing approximate vectors in the graph norm of a multiplication operator, and the Fourier transform preserves Schwartz space. Their unitary exponentials act explicitly by

$$U(a)f(x) = e^{iax}f(x), \quad V(b)f(x) = f(x + \hbar b), \quad a, b \in \mathbb{R}. \quad (13.3)$$

These assertions also follow from the multiplication spectral calculus and the Fourier transform, respectively; neither requires BCH. The functions (13.3) are unitary by a change of variables and strongly continuous by density of smooth compactly supported functions and dominated convergence.

Theorem 13.1 (Weyl and symmetric exponential identities). *In this representation,*

$$e^{iaQ}e^{ibP} = e^{i(aQ+bP-ab\hbar I/2)}, \quad (13.4)$$

$$e^{i(aQ+bP)} = e^{iaQ/2}e^{ibP}e^{iaQ/2}, \quad (13.5)$$

$$U(a)V(b) = e^{-iab\hbar}V(b)U(a). \quad (13.6)$$

More generally, with $\mathcal{W}(a, b) = e^{i(aQ+bP)}$,

$$\mathcal{W}(a, b)\mathcal{W}(c, d) = e^{i\hbar(bc-ad)/2}\mathcal{W}(a+c, b+d). \quad (13.7)$$

Proof. Define the unitary operator

$$\mathcal{W}(a, b)f(x) = e^{ia(x+\hbar b/2)}f(x+\hbar b). \quad (13.8)$$

For fixed a, b , the operators $\mathcal{W}(ta, tb)$ form a strongly continuous one-parameter group: direct multiplication of the phases proves the group law on this line. Its derivative at $t = 0$ on Schwartz functions is $i(aQ + bP)$. To identify its self-adjoint generator without ambiguity, for $b \neq 0$ multiplication by $e^{iax^2/(2b\hbar)}$, denoted T , preserves Schwartz space and satisfies

$$T^{-1}(bP)T = aQ + bP.$$

Thus this differential operator has the self-adjoint realization $T^{-1}(bP)T$, and direct conjugation of the translations gives (13.8). For $b = 0$ it is the multiplication operator aQ . This proves the claimed identification with the exponential.

Now $U(a)V(b)f = e^{iax}f(x+\hbar b)$, which is $e^{-iab\hbar/2}\mathcal{W}(a, b)f$. Multiplying $U(a/2)V(b)U(a/2)$ gives precisely (13.8). Comparing $U(a)V(b)$ with $V(b)U(a)$ proves (13.6). Finally multiply two instances of (13.8); after subtracting the phase of $\mathcal{W}(a+c, b+d)$, the remaining constant phase is $\hbar(bc-ad)/2$. This proves (13.7) on all of $L^2(\mathbb{R})$. $\square$

The formal BCH calculation agrees: for $A = iaQ$ and $B = ibP$, $[A, B] = -iab\hbar I$ is central, so only its one-half term remains. The preceding direct proof establishes that this formal calculation is legitimate for the chosen self-adjoint realizations.

For any finite list (a_j, b_j) , induction in (13.7) yields the complete finite-factor formula

$$\prod_{j=1}^m \mathcal{W}(a_j, b_j) = \exp\left(\frac{i\hbar}{2} \sum_{j < k} (b_j a_k - a_j b_k)\right) \mathcal{W}\left(\sum_j a_j, \sum_j b_j\right). \quad (13.9)$$

This is (9.4) in its unitary Heisenberg representation, with no omitted higher-order terms.

13.2 Why the infinitesimal commutator is not enough

Proposition 13.2. *A relation $[Q, P] = iI$ on a dense common invariant domain, even for self-adjoint Q, P , does not by itself imply the Weyl relations for all real parameters.*

Proof. Work on $L^2(0, 1)$. Let Q be multiplication by x , a bounded self-adjoint operator, and let $P = -i d/dx$ with periodic self-adjoint domain $\{f \in H^1(0, 1) : f(0) = f(1)\}$. Its Fourier basis $e^{2\pi i n x}$ has eigenvalues $2\pi n$; equivalently it is the self-adjoint diagonal operator with these real eigenvalues. The dense subspace $C_c^\infty(0, 1)$ is invariant under both operators and has $[Q, P] = iI$ by differentiation. But $e^{iP} = I$, since $e^{2\pi i n} = 1$ on every Fourier basis vector. The Weyl relation with parameter $b = 1$ would imply $e^{iaQ} = e^{-ia}e^{iaQ}$ for every real a , which is false unless $a \in 2\pi\mathbb{Z}$. Thus the dense-domain commutator is insufficient. The subspace used here is not a core for the periodic derivative; that distinction is exactly one of the domain issues hidden by purely formal notation. $\square$

There cannot even be bounded operators A, B with $[A, B] = cI$ for $c \neq 0$. Indeed induction gives $[A, B^n] = ncB^{n-1}$, so

$$n|c| \|B^{n-1}\| \leq 2 \|A\| \|B^n\|.$$

Iterating implies $\|B^n\| \geq n!|c|^n \|I\| / (2\|A\|)^n$, contrary to $\|B^n\| \leq \|B\|^n$ as $n \rightarrow \infty$. This explains why the bounded-operator convergence theorems cannot simply be applied to the canonical position and momentum pair.

13.3 Creation and annihilation in Bargmann–Fock space

Let $\mathcal{F}$ be the Hilbert space of entire functions with norm

$$\|F\|_{\mathcal{F}}^2 = \frac{1}{\pi} \int_{\mathbb{C}} |F(z)|^2 e^{-|z|^2} d^2z. \quad (13.10)$$

Polar integration gives $\langle z^m, z^n \rangle = \delta_{mn} n!$: the angular integral is zero for $m \neq n$, and the radial integral for $m = n$ is $n!$. Consequently

$$|n\rangle = \frac{z^n}{\sqrt{n!}}, \quad n \geq 0, \quad (13.11)$$

is an orthonormal basis. Completeness follows by expanding an entire function in its Taylor series, integrating first on circles, and then using monotone convergence for the sum of squared coefficient norms. In particular polynomials are dense. On this dense polynomial domain define

$$a = \partial_z, \quad a^\dagger = z, \quad [a, a^\dagger] = I, \quad [a^\dagger, a] = -I. \quad (13.12)$$

We take the Hilbert inner product to be linear in its second argument. The basis relations $a|n\rangle = \sqrt{n}|n-1\rangle$ and $a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$ prove that the two operators are adjoint there in the usual densely defined sense.

Theorem 13.3 (Displacement and normal ordering). *For $v \in \mathbb{C}$, the displacement operator has the explicit unitary action*

$$(D(v)F)(z) = e^{-|v|^2/2} e^{vz} F(z - \bar{v}). \quad (13.13)$$

It is the unitary exponential generated by $va^\dagger - \bar{v}a$, and on polynomials it satisfies the normal ordering identity

$$D(v) = e^{va^\dagger - \bar{v}a} = e^{va^\dagger} e^{-\bar{v}a} e^{-|v|^2/2}. \quad (13.14)$$

The right-hand side extends by continuity from that domain to the bounded operator $D(v)$; it is not a claim that both unbounded factors can be composed on every vector without a domain choice. For all $u, v \in \mathbb{C}$,

$$D(v)D(u) = \exp\left(\frac{v\bar{u} - u\bar{v}}{2}\right) D(v+u). \quad (13.15)$$

Proof. In (13.10), substitute $z = w + \bar{v}$ in the norm of (13.13). The total real exponent simplifies to

$$-|v|^2 + 2\operatorname{Re}(v(w + \bar{v})) - |w + \bar{v}|^2 = -|w|^2.$$

Thus $D(v)$ is isometric. Direct substitution gives $D(-v)$ as its inverse, so it is unitary. Multiplying its explicit actions yields

$$D(v)D(u)F(z) = e^{-(|v|^2 + |u|^2)/2 - u\bar{v}} e^{(v+u)z} F(z - \bar{v} - \bar{u}).$$

Comparison with $D(v+u)$ proves (13.15), including its sign and factor $1/2$.

For a fixed v , $D(tv)$ is a strongly continuous one-parameter unitary group for real t ; its group law follows from (13.15) because the central phase vanishes on this line. Strong continuity follows first for polynomials from the explicit formula and Gaussian integrability, and then for every vector by density and unitarity. Its derivative at zero on polynomials is $va^\dagger - \bar{v}a$. The exponential notation in (13.14) denotes this skew-adjoint generator. There is no ambiguity about its closure: on a basis vector the k th power T^k , $T = va^\dagger - \bar{v}a$, satisfies

$$\|T^k|n\rangle\| \leq (2|v|)^k \sqrt{\frac{(n+k)!}{n!}}. \quad (13.16)$$

This follows by noting that at step j all occupied levels are at most $n+j$, and each of $a, a^\dagger$ has norm at most the square root of the next level on that finite-dimensional subspace. Thus $\sum_k \|T^k F\| |t|^k/k!$ converges for every polynomial F and every t . To check uniqueness explicitly, if a vector η were in a nonzero deficiency space for the skew-symmetric operator T , it would satisfy $T^*\eta = \eta$ or $T^*\eta = -\eta$. For polynomials the identity $D(tv)F = \sum_{k \geq 0} t^k T^k F/k!$ follows from the integral Taylor remainder: its norm is at most $|t|^{k+1} \|T^{k+1} F\|/(k+1)!$, which tends to zero by (13.16). Hence $\langle \eta, D(tv)F \rangle$ would equal $e^t \langle \eta, F \rangle$ or $e^{-t} \langle \eta, F \rangle$. Unitarity bounds this scalar for all positive and negative real t , forcing $\langle \eta, F \rangle = 0$ for all polynomials and hence $\eta = 0$. To finish the operator argument explicitly, let S be the closure of T , which exists because T is densely defined and skew-symmetric. The identity $\|(I \pm S)u\|^2 = \|u\|^2 + \|Su\|^2$ shows that both ranges are closed. Their orthogonal complements are the two vanishing deficiency kernels, so both ranges are the entire Hilbert space. For $\eta \in \text{Dom}(S^*)$, solve $(I - S)u = (I + S^*)\eta$. Since $S^*u = -Su$, the difference $\eta - u$ lies in $\ker(I + S^*) = 0$. Thus $\eta = u \in \text{Dom}(S)$, proving $S^* = -S$. The explicit unitary group therefore supplies this unique skew-adjoint closure, with polynomials as a core.

Finally, on a polynomial F , $e^{-\bar{v}a}F = F(z - \bar{v})$ by the finite Taylor formula, while $e^{va^\dagger}$ multiplies by e^{vz} . Their ordered product with the scalar factor is exactly (13.13). This proves the normal ordering assertion on the stated domain and hence its bounded extension. $\square$

The commutator of the two generators is

$$[va^\dagger - \bar{v}a, ua^\dagger - \bar{u}a] = (v\bar{u} - u\bar{v})I.$$

It is central and purely imaginary. Thus (13.15) is also exactly the central-commutator BCH identity, now justified by a representation-level proof rather than by manipulating undefined operator products.

13.4 All matrix elements and finite products

Every finite displacement product has the closed expression

$$\prod_{j=1}^m D(v_j) = \exp\left(\frac{1}{2} \sum_{j < k} (v_j \bar{v}_k - v_k \bar{v}_j)\right) D\left(\sum_j v_j\right), \quad (13.17)$$

by induction in (13.15). Acting on the vacuum gives the full coherent-state series

$$|v\rangle = D(v)|0\rangle = e^{-|v|^2/2} \sum_{n \geq 0} \frac{v^n}{\sqrt{n!}} |n\rangle, \quad \langle u|v\rangle = e^{-(|u|^2 + |v|^2)/2 + \bar{u}v}. \quad (13.18)$$

The first formula is the exponential series for e^{vz} , and the second follows by the orthonormal basis expansion and the scalar exponential series. The norm at $u = v$ is one. More generally all number-state matrix elements are the finite sums

$$\langle m|D(v)|n\rangle = e^{-|v|^2/2}\sqrt{m!n!}\sum_{k=0}^{\min(m,n)}\frac{v^{m-k}(-\bar{v})^{n-k}}{(m-k)!(n-k)!k!}. \quad (13.19)$$

To derive this, apply the finite lowering series $e^{-\bar{v}a}$ to $|n\rangle$, leaving level k with coefficient $(-\bar{v})^{n-k}\sqrt{n!/k!}/(n-k)!$. The raising series $e^{va^\dagger}$ then reaches level m with coefficient $v^{m-k}\sqrt{m!/k!}/(m-k)!$. Multiplying and summing over the admissible k proves (13.19). These formulas provide all terms of the oscillator applications; no asymptotic truncation remains.

A An elementary PBW proof and the enveloping Hopf algebra

This appendix supplies the injectivity statement used in the universal-algebra discussion. It also proves that the Hopf structure exists on an arbitrary universal enveloping algebra, not only on the free associative algebra.

Let $\mathfrak{h}$ be a Lie algebra with an ordered basis $\{e_i\}_{i \in I}$. Its enveloping algebra is the quotient of the tensor algebra on its underlying vector space by the relations

$$uv - vu = [u, v] \quad (u, v \in \mathfrak{h}). \quad (A.1)$$

It has the following universal property: every Lie homomorphism from $\mathfrak{h}$ into the commutator algebra of an associative algebra extends uniquely to an associative homomorphism from $U(\mathfrak{h})$. Indeed it first extends uniquely to the tensor algebra, and (A.1) is exactly the condition that the resulting map annihilate the quotient ideal.

Theorem A.1 (Poincaré–Birkhoff–Witt). *The ordered monomials $e_{i_1} \cdots e_{i_m}$ with $i_1 \leq \cdots \leq i_m$, including the empty monomial, form a vector-space basis of $U(\mathfrak{h})$. In particular the natural map $\mathfrak{h} \rightarrow U(\mathfrak{h})$ is injective.*

Proof. Replace an adjacent inverted pair by

$$e_j e_i \mapsto e_i e_j + [e_j, e_i] \quad (j > i), \quad (A.2)$$

expanding the bracket in the given basis. Order words by their length first and their number of inversions second. The swapped word has the same length and one fewer inversion, whereas every bracket term has smaller length. This is a well-founded decreasing measure, so repeated rewriting terminates in a linear combination of ordered words. It proves spanning.

To prove uniqueness of the result, consider two first reduction choices. Reductions on disjoint adjacent pairs commute. The only overlapping ambiguity is a word cba with three strictly decreasing basis indices $c > b > a$. Reducing the left inverted pair first and then ordering the remaining triple gives

$$abc + [b, a]c + b[c, a] + [c, b]a.$$

Reducing the right inverted pair first instead gives

$$abc + a[c, b] + [c, a]b + c[b, a].$$

The difference, using the length-two relations, reduces to

$$[[b, a], c] + [b, [c, a]] + [[c, b], a] = 0$$

by Jacobi. Each bracket-expanded term has smaller length than the original triple. With a prefix and suffix attached, it still has smaller length than the original word. Induction on the decreasing measure therefore already guarantees a unique normal form for these differences.

More explicitly, suppose every strictly smaller word has a unique normal form. For two reductions of the current word, the disjoint case has a common next reduction, while the overlapping case just computed has difference with zero normal form among smaller words. All subsequent choices act on smaller words, so their normal forms agree by induction. This proves uniqueness for the current word, and hence for every word and by linearity for every polynomial. The resulting normal-form map annihilates each contextual relation (A.2), hence its two-sided ideal. Conversely a polynomial and its normal form differ by that ideal. Therefore the quotient is exactly the vector space freely spanned by the ordered words. In particular the length-one basis vectors remain linearly independent, proving injectivity. $\square$

No matrix realization of $\mathfrak{h}$ and no BCH theorem was used in this proof, so the earlier use of PBW is not circular.

On the tensor algebra set $\Delta u = u \otimes 1 + 1 \otimes u$, $\varepsilon(u) = 0$, and $S(u) = -u$ for $u \in \mathfrak{h}$, extending Δ, ε multiplicatively and S antimultiplicatively. For a defining relation $r = uv - vu - [u, v]$, direct expansion gives

$$\Delta r = r \otimes 1 + 1 \otimes r, \quad \varepsilon(r) = 0, \quad S(r) = -r.$$

Consequently these operations descend to the quotient $U(\mathfrak{h})$. The coproduct is coassociative on generators and therefore on the whole algebra; the counit and antipode identities proved for tensor algebras descend as well. This is the natural enveloping Hopf algebra structure asserted on the reference page.

To see concretely why an infinite-series coproduct needs a completed tensor product, restrict to one generator and consider $F(X) = \sum_{n \geq 0} X^n$. Its formal coproduct has coefficient matrix

$$[X^p \otimes X^q] \Delta F = \binom{p+q}{p} \quad (p, q \geq 0).$$

The leading $(N+1) \times (N+1)$ matrix is PP^T , where $P_{pk} = \binom{p}{k}$ for $k \leq p$ and is zero otherwise. The identity $\binom{p+q}{p} = \sum_k \binom{p}{k} \binom{q}{k}$ is Vandermonde's identity, obtained by comparing coefficients in $(1+t)^p(1+t)^q$. Since P is triangular with diagonal 1, every such matrix has determinant 1. Its rank is therefore unbounded with N . But the coefficient matrix of a finite sum of separated tensors $\sum_{j=1}^m A_j(X) \otimes B_j(X)$ has rank at most m . Thus ΔF is not in the ordinary algebraic tensor product, while it is in the completed tensor product used in section 3.

B Complete word certificates for the displayed fifth and sixth degrees

In the following tables an entry is simultaneously the coefficient $c(w)$ computed by the finite cut formula (2.5) and the coefficient obtained by expanding (6.5) or (6.6). Thus the tables specify every coordinate needed for the polynomial identities, including zeros. Right-nested brackets are expanded by the rule $[av]_R = a[v]_R - [v]_R a$, starting with $[a]_R = a$. For example, the coefficient of $X^4 Y$ in Z_5 is $-1/720$, and that of $X^5 Y$ in Z_6 is zero, as the single- Y Bernoulli coefficients also predict.

Degree 5 word coefficients

Word	Coefficient	Word	Coefficient
XXXXX	0	YXXXX	$-\frac{1}{720}$
XXXXY	$-\frac{1}{720}$	YXXXY	$\frac{1}{180}$
XXXYX	$\frac{1}{180}$	YXXYX	$-\frac{1}{120}$
XXXYY	$\frac{1}{180}$	YXXYY	$-\frac{1}{120}$
XXYXX	$-\frac{1}{120}$	YXYXX	$-\frac{1}{120}$
XXYXY	$-\frac{1}{120}$	YXYXY	$\frac{30}{120}$
XXYYX	$-\frac{1}{120}$	YXYYX	$-\frac{1}{120}$
XXYYY	$\frac{1}{180}$	YXYYY	$\frac{1}{180}$
XYXXX	$\frac{1}{180}$	YYXXX	$\frac{1}{180}$
XYXXY	$-\frac{1}{120}$	YYXXY	$-\frac{1}{120}$
XYXYX	$\frac{30}{120}$	YYXYX	$-\frac{1}{120}$
XYXYY	$-\frac{1}{120}$	YYXYY	$-\frac{1}{120}$
XYYYX	$-\frac{1}{120}$	YYYXX	$\frac{1}{180}$
XYYYX	$-\frac{1}{120}$	YYYXY	$\frac{1}{180}$
XYYYX	$\frac{1}{180}$	YYYYX	$-\frac{1}{720}$
XYYYY	$-\frac{1}{720}$	YYYYY	0

Degree 6 word coefficients

Word	Coefficient	Word	Coefficient	Word	Coefficient	Word	Coefficient
XXXXXX	0	YXXXXX	0	YXXXXX	0	YYXXXX	$\frac{1}{1440}$
XXXXXY	0	YXXXXY	$\frac{1}{360}$	YXXXXY	0	YYXXXY	0
XXXXYX	0	YXXXYX	0	YXXXYX	$-\frac{1}{360}$	YYXXYX	$\frac{1}{240}$
XXXXYY	$-\frac{1}{1440}$	YXXXYX	$-\frac{1}{240}$	YXXXYX	0	YYXXYY	0
XXXXYX	0	YXXYXX	0	YXXYXX	$\frac{1}{240}$	YYXYXX	$\frac{1}{240}$
XXXXYX	$\frac{1}{360}$	YXXYXY	$\frac{1}{60}$	YXXYXY	0	YYXYXY	0
XXXYYX	0	YXXYYX	0	YXXYYX	$\frac{1}{240}$	YYXYXX	$\frac{1}{240}$
XXXYYY	$\frac{1}{360}$	YXXYYY	$\frac{1}{360}$	YXXYYY	0	YYXYYY	0
XXYXXX	0	YXYXXX	0	YXYXXX	$-\frac{1}{360}$	YYYXXX	$-\frac{1}{360}$
XXYXXY	$-\frac{1}{240}$	YXYXXY	$-\frac{1}{240}$	YXYXXY	0	YYYXXY	0
XXYXYX	0	YXYXYX	0	YXYXYX	$-\frac{1}{60}$	YYYXYX	$-\frac{1}{360}$
XXYXYX	$-\frac{1}{240}$	YXYXYX	$-\frac{1}{240}$	YXYXYX	0	YYYXYX	0
XXYXXY	0	YXYXXY	0	YXYXXY	$\frac{1}{240}$	YYYYXX	$\frac{1}{1440}$
XXYXYX	$-\frac{1}{240}$	YXYXYX	$\frac{1}{360}$	YXYXYX	0	YYYYXY	0
XXYYYX	0	YXYYYX	0	YXYYYX	$-\frac{1}{360}$	YYYYXX	0
XXYYYY	$-\frac{1}{1440}$	YXYYYY	0	YXYYYY	0	YYYYYY	0

The general formula is not inferred from these tables; it was proved in [theorems 2.2](#) and [3.6](#). The tables verify the specific low-degree displays and make every arithmetic step independently reproducible. The accompanying TSV and JSON files continue the complete nonzero associative coefficients through degree ten. Their size is modest because repeated word coefficients have already been combined.

C Reproducible exact computations

The program `code/verify_bch.py` uses Python's standard library only, with exact rational numbers throughout. A polynomial is a dictionary from words to rational coefficients, and products concatenate words while discarding degrees above a chosen cutoff. The empty word represents the identity. This works in the finite quotient of the free associative algebra by words of degree greater than the cutoff; no sampling of numerical matrices is involved.

The core logarithmic calculation is simply

```
E = exp(X) * exp(Y) # all operations truncated by degree
U = E - 1
Z = sum((-1)**(k-1) * U**k / k for k in range(1, N+1))
```

The actual implementation retains exact fractions and sparse word dictionaries. It compares this result with the independent Bernoulli differential recurrence, the right Dynkin projection, and the independently enumerated cut formula. Three recursive Zassenhaus constructions are compared with explicit enumeration of the finite weighted-tree formula. Their products are then multiplied out and compared with e^{X+Y} .

The included run used cutoff $N = 10$ and passed all 20 named checks. Most checks run through degree ten; two independent exhaustive checks, the word-cut enumeration and unshuffle coproduct, run through degree eight, and three-letter associativity through degree six. The page’s finite displays are checked through their full displayed degrees, six for BCH and four for Zassenhaus. These exact finite checks do not claim machine verification of the infinite-order or analytic proofs.

Degree	Nonzero BCH words	Nonzero C_n words	Weighted trees
1	2	0	0
2	2	2	1
3	6	6	1
4	4	12	2
5	30	30	2
6	28	42	5
7	126	126	6
8	124	234	16
9	390	510	22
10	388	954	54

These counts concern the associative word expansion, not dimensions of homogeneous free Lie spaces and not the number of terms in a chosen Hall basis. The weighted-tree column counts trees in [theorem 10.3](#) before their associative terms are combined.

To reproduce the results from the archive’s top-level directory:

```
python3 code/verify_bch.py --degree 10 --output data
latexmk -pdf -interaction=nonstopmode -halt-on-error bch_complete.tex
```

Python 3.10 or later is suitable. The verifier permits degrees between six and twelve; increasing the cutoff increases the amount of exact arithmetic substantially. The LaTeX source uses standard packages and does not require bibliography software or shell escape. When `latexmk` is unavailable, run `pdflatex` twice (or until cross-reference warnings disappear).

D Coverage index and necessary corrections

This index uses the pinned reference-page version stated in [section 1](#). Every displayed mathematical formula and every substantive mathematical assertion on that page is covered below. A corrected assertion is identified as such rather than presented as though an incorrect unrestricted version had been proved.

Reference-page formula or assertion	Proof and interpretation here
Product of exponentials has a formal Lie-series logarithm; leading BCH terms	Lemma 2.1 and corollary 3.5 ; (6.1)–(6.3).
Only commutators and their nestings occur beyond $X + Y$	Corollary 3.5 , including the ideal generated by $[X, Y]$.
Small elements give a convergent BCH series	Theorem 7.1 and corollary 7.2 ; the Banach Lie-algebra bound also covers all finite-dimensional Lie algebras.
Every group element near the identity is an exponential; local multiplication is Lie-algebraic	Theorem 8.1 ; intrinsic differential and inverse function theorem.
Lie correspondence and integration of representations as applications	Theorems 8.2 and 8.3 ; the latter includes the simply connected hypothesis for global integration.
Dynkin’s multiple sum, its factorial and degree denominators	Theorem 3.6 ; every word coefficient in theorem 2.2 .
Right-nesting notation and zero final blocks	(1.2); the final assertion of theorem 3.6 .
Every BCH term through degree six	(6.1)–(6.6); derivation in section 6 and full certificates in section B .
$X \leftrightarrow Y$ symmetry in alternating degrees	(6.8)–(6.9), proved by inverse and logarithm uniqueness.
Poincaré integral formula	Theorem 5.2 ; derived from the differential equation rather than assumed as an external identity.
$\psi(x) = x \log x / (x - 1)$ and its complete Taylor series	(5.3); valid near $x = 1$ or formally.
$\psi(e^y)$ and normalized Bernoulli coefficients in the note	(5.4), proposition 4.3 ; convention $B_1^+ = 1/2$ is explicit.
Matrix tangent Lie algebra, commutator, and matrix exponential	Section 8 ; the flow equation identifies the ordinary exponential.
Associative logarithmic block sum	(2.3); absolute convergence in theorem 7.1 .
Conditions involving $\log 2$, including the note on Hilbert–Schmidt norm	Section 7 ; submultiplicativity is proved and the extra bound on a constructed Z is explained as unnecessary.
Associative terms of degrees one through four	(6.7), obtained by direct bracket expansion.
The trace logarithm identity	Proposition 7.3 ; local BCH branch required, with the exact $2\pi i\mathbb{Z}$ ambiguity for other logarithms.
The $\mathfrak{sl}_2(\mathbb{C})$ example and its nonconvergence	Theorem 7.4 ; all matrix logarithms are explicitly classified, and Abel’s argument rules out even conditional BCH convergence.
Nonisomorphic Lie groups may have the same Lie algebra	The explicit $\mathbb{R}$ and circle example after theorem 8.1 .
Sufficient norm bound $(\log 2)/2$	(7.2); it controls the unreduced nested multiple sum.
Commuting and central-commutator special cases; no smallness restriction in the latter	Theorem 9.1 ; a global differential proof in the stated bounded/group settings.
Expansion linear in Y , including the hyperbolic-cotangent form	(5.7); all higher powers are supplied by proposition 5.3 .
$[X, Y] = sY$ eliminates terms with two or more Y ’s	The Lie-monomial induction in theorem 9.2 .
Closed eigenvector BCH coefficient, resonance exclusion, and $s = 0$ limit	(9.8); both removable zero and genuine nonzero resonances are proved.
Eigenvector braiding and adjoint dilation	(9.6) and its proof; multiplying by e^{-X} gives the displayed dilation.
Formal exponential/logarithm bijection	Lemma 2.1 .
Coproduct of the generators and extension to series	(3.1)–(3.2); the tensor product is completed. Section A gives a counterexample to an uncompleted interpretation.

Reference-page formula or assertion	Proof and interpretation here
Exponential is grouplike iff exponent is primitive; grouplikes form a group	Lemma 3.2 , with constant term 1 in the definition of grouplike.
Friedrichs' characterization and the primitive-logarithm existence argument	Lemma 3.3 , theorem 3.4 , and corollary 3.5 .
Enveloping algebra of the free Lie algebra is the free associative algebra; natural Hopf structure	End of section 3 and section A ; the completion is specified explicitly.
Zassenhaus product and every displayed exponent	Theorem 10.1 ; (10.4)–(10.6).
All higher Zassenhaus exponents are homogeneous Lie polynomials; recursive determination	Theorem 10.1 ; residual, differential, and forward algorithms; finite all-terms formula in theorem 10.3 .
Alternative Zassenhaus notation starting at $n = 1$	The paragraph after theorem 10.1 supplies the necessary convention $C_1 = Y$.
Suzuki–Trotter consequence	Theorems 11.1 and 11.2 , including all-order local logarithmic error series.
$\text{Ad}_{e^X} = e^{\text{ad}_X}$, complete iterated-commutator sum, and its linear differential equation	Theorem 4.1 and (4.2).
Conjugation of e^Y and general braiding identity	(4.3)–(4.4).
The central-commutator application obtained by differentiating $e^{sX}e^{sY}$	Proof of theorem 9.1 , including its global initial-value argument.
Infinitesimal $e^{-X}de^X$ and its component terms	(12.2)–(12.3); all component orders in (12.6).
$M_b^a = X^i f_{ib}^a$, series for W , and quotient notation	(12.4); proposition 12.1 explains the removable functional calculus and singularity of M .
The page's statement that M is a vielbein	<i>Corrected:</i> W is the coframe matrix and W^{-1} its dual frame, where defined; M is always singular. (12.9)–(12.10); an arbitrary map needs its Jacobian. Full series in theorem 12.2 .
Pullback tensor $g_{ij} = W_i^a W_j^b B_{ab}$	<i>Qualified:</i> (12.11) and its invariance proof; nondegeneracy and pullback hypotheses, with explicit counterexamples, in section 12 .
Killing form and assertions that the pullback is a metric	(13.1)–(13.2), proved on Schwartz space.
Canonical position–momentum commutator	Theorem 13.1 ; direct unitary actions establish the domain-safe meaning. Proposition 13.2 explains the limitation of a bare infinitesimal relation.
Exponentiated position–momentum and symmetric identities	(13.12) and theorem 13.3 ; explicitly stated dense domain and bounded extension.
$[a^\dagger, a] = -I$ and displacement normal ordering	(13.15); all finite products and number-state matrix elements in (13.17)–(13.19).
Product of two displacements and its central phase	(9.5), proposition 8.4 , and the displayed central generator commutator after theorem 13.3 .
Heisenberg nilpotence underlying those applications	

The genuinely all-terms outputs can be located without scanning the index: every BCH coefficient is in (2.5) and (3.8); every power of Y in (5.6) and (5.9); every Zassenhaus exponent in (10.14) and (10.15); every Maurer–Cartan component in (12.4)–(12.6); every invariant metric coefficient in (12.14); and every oscillator matrix element in (13.19). These statements specify coefficients at arbitrary order, not just an extensive initial truncation.

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